Does Austerity Redistribute?

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Montevideo - Uruguay

Octubre de 2026

FACULTAD DE CIENCIAS ECONÓMICAS Y DE ADMINISTRACIÓN

Ciencias Sociales Universidad de la República URUGUAY

UNIVERSIDAD DE LA REPÚBLICA URUGUAY

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Tesis de Maestría presentada al Programa de Maestría en Economía de la Facultad de Ciencias Económicas y de Administración y la Facultad de Ciencias Sociales, Universidad de la República, como parte de los requisitos para la obtención del título de Magíster en Economía para la investigación.

Directores de tesis:

Ronald Miranda y Leonel Muinelo

Montevideo - Uruguay

Octubre de 2026

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Abstract

Does fiscal austerity change who bears the burden of adjustment? This thesis studies the dynamic distributional effects of fiscal consolidations for 30 economies in the OECD and in Latin America and the Caribbean over 1996–2019, identifying episodes through the narrative approach of Adler et al. (2024). Departing from the standard regional comparison, this thesis lets the response vary continuously with quality of government (QoG), a composite measure of state capacity. Pooling all episodes, neither the market nor the disposable Gini responds detectably; redistribution, the gap between the two, does respond. Once QoG is allowed to condition the response, a clear pattern emerges: in high-QoG economies absolute redistribution widens by roughly 0.23 Gini points per percentage point of GDP of consolidation by the third year after the shock, a response that is robust over the early-to-medium run but not at longer horizons, while low-QoG economies show no response in the baseline specification. The effect is stronger for tax-based than for expenditure-based adjustments, consistent with a progressivity channel that requires administrative capacity to operate. Austerity has modest and short-lived redistributive effects on average, but these effects are substantially stronger and more persistent where the state is already capable of redistributing.

Keywords: Fiscal consolidation; Austerity; Income redistribution; State capacity; Quality of government; Local projections.

JEL Classification: E62; H23; H50; D31; O23.

Resumen

¿La austeridad fiscal modifica quién soporta la carga del ajuste? Esta tesis estudia los efectos distributivos dinámicos de las consolidaciones fiscales para 30 economías de la OCDE y de América Latina y el Caribe durante el período 1996–2019, identificando los episodios mediante el enfoque narrativo de Adler et al. (2024). A diferencia de la literatura, que compara por región, esta tesis permite que la respuesta varíe de forma continua con la calidad de gobierno (QoG), una medida compuesta de la capacidad estatal. En la estimación agrupada, ni el Gini de mercado ni el Gini disponible responden de forma detectable; la redistribución, la brecha entre ambos, sí lo hace. Al permitir que la QoG condicione la respuesta, emerge un patrón claro: en las economías con QoG alta, la redistribución absoluta se amplía en aproximadamente 0,23 puntos de Gini por cada punto porcentual del PIB de consolidación al tercer año, una respuesta robusta en el corto y mediano plazo pero no en horizontes largos, mientras que las economías con QoG baja no muestran respuesta en la especificación de base. El efecto es más fuerte para los ajustes basados en impuestos que para los basados en gasto, en consonancia con un canal de progresividad que requiere capacidad administrativa para operar. La austeridad tiene, en promedio, efectos redistributivos modestos y de corta duración, pero estos efectos son sustancialmente más fuertes y persistentes allí donde el Estado ya es capaz de redistribuir.

Palabras clave: Consolidación fiscal; Austeridad; Redistribución de ingreso; Capacidad estatal; Calidad de gobierno; Proyecciones locales.

Clasificación JEL: E62; H23; H50; D31; O23.

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1 Introduction

Austerity is never politically neutral. When governments cut spending or raise taxes to reduce public debt, they are not simply making technical adjustments to a balance sheet; they are making choices about who bears the burden of fiscal discipline. Yet the vast literature on fiscal consolidations has focused on their macroeconomic consequences, such as output losses, unemployment, and the pace of debt reduction. The distributional question (*who pays?*) has received comparatively less attention, and what evidence exists is largely confined to either advanced Organisation for Economic Co-operation and Development (OECD) economies or Latin America and the Caribbean (LAC) in isolation (Ball et al., 2013; Heimberger, 2020; Komatsuzaki, 2024). Whether the inequality effects of austerity are structurally different across levels of development, and whether findings from one region can be generalized to the other, remains an open question. This thesis argues that this question should be central.

Recent history offers no shortage of such episodes. Ireland consolidated for six consecutive years between 2009 and 2014 in the aftermath of its banking crisis; the United Kingdom sustained a comparable adjustment from 2010 to 2015; Spain and Portugal compressed their deficits through the euro-area sovereign debt crisis over 2010–2014. Latin America's experience is more recent still. Argentina tightened in 2018–2019, and Brazil adjusted in 2015 and 2017 in the midst of its recession.[1] Each was undertaken under mounting debt pressure, but the fiscal machinery through which the adjustment passed differed sharply across them. Studying either group in isolation cannot establish whether that difference matters for who ends up paying.

Fiscal consolidations are understood as adjustments made with the purpose of reducing debt as a share of GDP, in which "austerity" measures are taken against an excessive accumulation of debt[2] as a result of policy oversights and political distortions (Alesina and Passalacqua, 2016). These episodes frequently follow economic crises, given the key role of fiscal policy in response to downturns. Public debt rises as spending increases and revenue falls, prompting governments concerned about fiscal sustainability to implement consolidation measures through combinations of tax hikes and spending cuts (Furceri et al., 2015).[3]

These episodes are not a closed chapter. Public debt across advanced and developing economies reached postwar highs after the Great Recession and the COVID-19 pandemic (International Monetary Fund, 2020), and governments must now rebuild fiscal buffers against weaker growth prospects and higher borrowing costs (International Monetary Fund, 2024, 2025). The societies absorbing this coming round of adjustment are more unequal than those that absorbed the last, after four decades of rising within-country inequality (Atkinson et al., 2011; Chancel et al., 2022) interrupted only briefly by pandemic-era transfers (Meyer et al., 2025). How state capacity shapes the distributional consequences of austerity is therefore a live question, even though the episodes analyzed here end in 2019.

The contrast between OECD and LAC countries shown in Figure 1 offers a first, intuitive entry point for thinking about why the distributional consequences of austerity might differ across settings. Advanced OECD economies are characterized by larger government sectors, more progressive tax structures, and broader social protection systems, which suggests that the redistributive channels through which consolidation operates (cuts to transfers, reductions

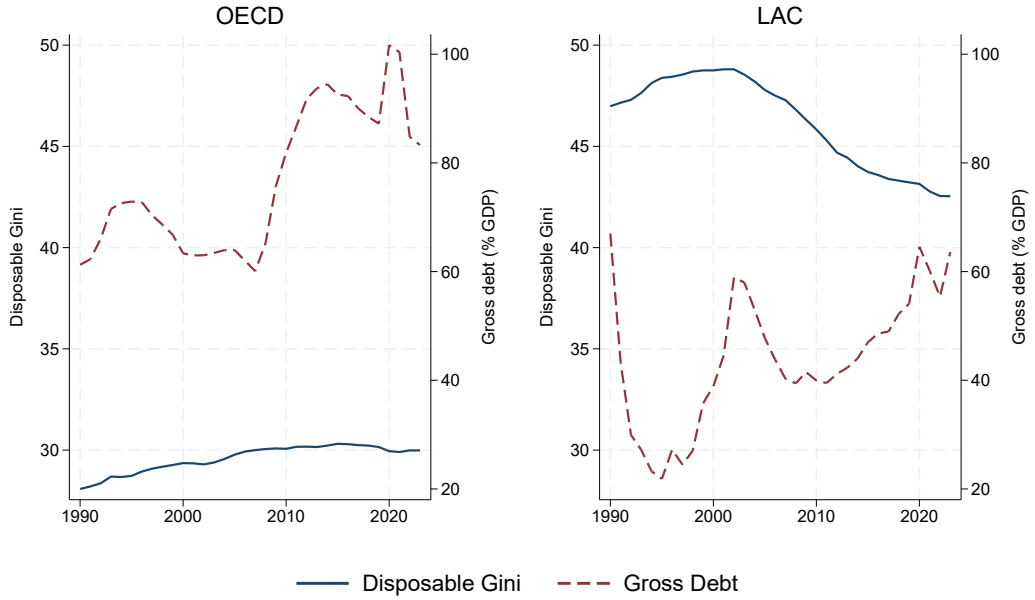
[1] All of these episodes appear in the sample analyzed below and are listed in Table 8 (Appendix B.2).

[2] Excessive from the standpoint of a social planner.

[3] However, a reduction in the debt to GDP ratio does not always occur. On average, these consolidations do not have significant effects on that ratio, and their effectiveness is conditional on the macroeconomic context (Ando et al., 2025).

in public services, or increases in direct taxation) differ substantially from those in LAC. In the latter, tax systems are more regressive, automatic stabilizers are weaker, and the reach of contributory transfers is more limited. The depth of social protection in the OECD group matters directly here. Unemployment insurance, public pensions, and near-universal health coverage insure a large share of household income, and a consolidation reaches the disposable distribution through that apparatus.

Figure 1: Disposable Gini and gross debt average per region



Note: Disposable Gini and government gross debt average per region from 1990 to 2023. Disposable Gini reflects income inequality after taxes and government transfers. Gross debt as percentage of GDP. Both groups are restricted to the 30 countries in the analysis sample, listed in Table 8 in Appendix B.2. Correlations are 0.62 for OECD and -0.53 for LAC. Source: Own elaboration based on data from the Standardized World Income Inequality Database (SWIID) and World Bank.

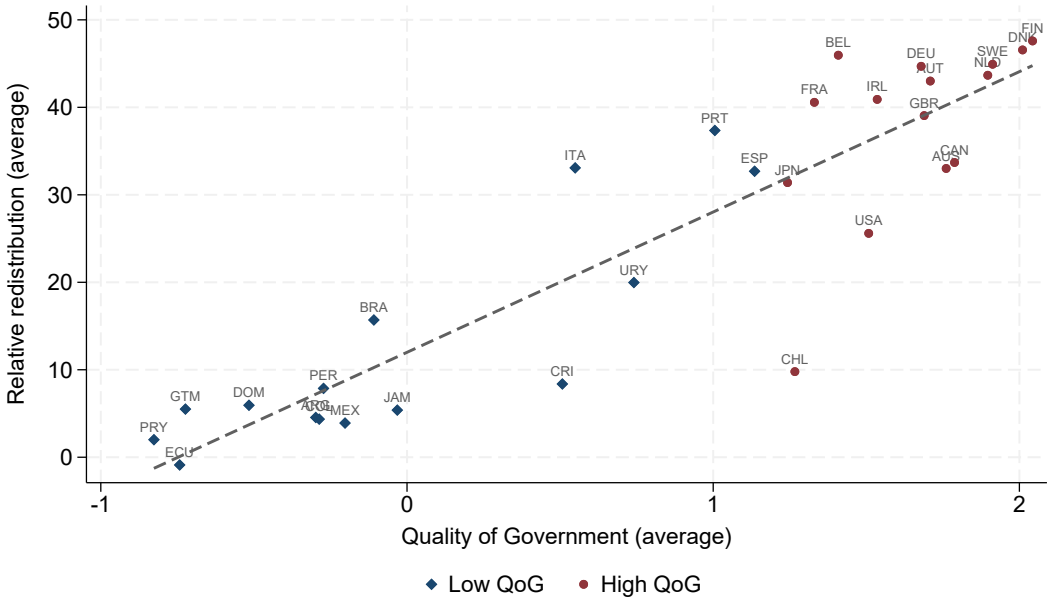
While the OECD–LAC dichotomy offers a useful descriptive frame, it is ultimately an imperfect basis for identifying the structural determinants of these differences. OECD membership is organized along a criterion of development and institutional convergence, whereas LAC is a geographic designation.[4] The two groups differ across many dimensions simultaneously: tax design, transfer coverage, automatic stabilizers, demographics, financial development, and trade structure, which makes it difficult to isolate which feature drives any heterogeneity that may be observed. The empirical analysis that follows therefore lets the response vary with the quality of government (QoG), a composite measure of state capacity spanning control of corruption, government effectiveness, rule of law, and regulatory quality, rather than with region. This grouping has direct theoretical content in the heterogeneous-household framework developed below. QoG proxies for the historical capacity of the state to tax broadly, administer transfers without leakage, and enforce compliance, and therefore for the breadth of the tax base, the coverage of transfers, and the size of the informal sector that sits outside the reach of both, the very

[4]The two labels are used throughout as sample groupings, not as statements of institutional membership. Chile, Colombia, Costa Rica, and Mexico belong to the OECD and are nonetheless grouped here as LAC, so the partition is development-based on one side and geographic on the other. The overlap is itself informative: it shows that the regional frame keeps together countries that differ sharply in state capacity, which is precisely what the QoG measure used below distinguishes.

channels through which fiscal consolidations alter the income distribution. The regional contrast is retained below only as descriptive motivation, and every estimation is organized around QoG instead.

Figure 2 shows this directly: across the sample, countries with a higher QoG exhibit systematically higher average redistribution. The association is cross-sectional, between average governance and average redistribution, and says nothing about how either responds to a fiscal shock. Establishing that requires conditioning on covariates, country fixed effects, and the size and timing of the consolidation.

Figure 2: QoG and redistribution across countries



Note: Country averages, over the 1996–2019 analysis sample, of QoG, the simple mean of the Worldwide Governance Indicators' estimates of control of corruption, government effectiveness, rule of law, and regulatory quality, against relative redistribution, the percent gap between the market and disposable Gini. The dashed line is a linear fit. Labels are ISO alpha-3 country codes (see Appendix B.2). Source: Own elaboration based on data from WGI and SWIID.

This thesis investigates what happens to inequality when governments tighten the fiscal belt, and whether the answer depends on *how* and *where* they do it. Specifically, it examines whether the distributional effects of fiscal consolidations differ between economies with high and low QoG, and whether consolidations driven by spending cuts produce different distributional outcomes than those driven by tax hikes once QoG is accounted for. To address the identification problem inherent in this literature, the analysis identifies consolidation episodes through the narrative approach rather than a statistical approach based on the cyclically adjusted primary balance (CAPB), and estimates their dynamic effects on market and disposable inequality, and on absolute and relative redistribution, using local projections on a pooled panel of 30 OECD and LAC economies over 1996–2019.

The answer that emerges is that a fiscal consolidation's distributional consequences run through redistribution rather than through changes in either Gini measure, neither of which moves detectably in the pooled sample. The redistributive response is modest and short-lived on average, but becomes stronger and more persistent in economies with high QoG. There, absolute redistribution rises by roughly 0.23 points per percentage point of gross domestic product (GDP) of consolidation by the third year after the shock, a response driven more strongly by tax-

based than by expenditure-based adjustment, while low-QoG economies show no such response in the baseline specification. This thesis contributes to the literature on fiscal consolidations and inequality in two main ways: by reframing the standard regional comparison between OECD and LAC economies around QoG, a structural measure of state capacity, rather than geography, as the relevant dimension of heterogeneity; and by showing that austerity's distributional consequences appear along the margin of redistribution rather than in the level of either Gini, and that this margin moves only where state capacity is high.

The remainder of this thesis is organized as follows. Section 2 reviews the literature on the macroeconomic and distributional effects of fiscal consolidation, on state capacity, and on fiscal incidence, and closes with the two research questions this thesis answers. Section 3 develops a heterogeneous-household model of taxation and transfers, adds to it a set of assumptions about how state fiscal capacity shapes each channel, and derives the two hypotheses tested empirically. Section 4 describes the data and the local-projections methodology, including the narrative identification of consolidation episodes. Section 5 presents the main results, moving from the pooled average effect through its composition to the QoG-conditional response that is the central finding of the thesis. Section 6 summarizes the robustness checks to which this result is subjected, detailed in full in Appendix A. Section 7 discusses the limitations of the empirical design, and Section 8 concludes with the policy implications of the results.

2 Literature review

The theoretical debate surrounding fiscal consolidation is defined by the tension between the standard Keynesian proposition and the potential for non-Keynesian expansionary effects, as Okwuokei (2014) discusses. While Keynesian theory suggests that reducing deficits is inherently contractionary due to lower aggregate demand, contemporary research highlights expansionary fiscal contractions driven by rational expectations (Alesina and Ardagna, 2010). These occur primarily through the wealth effect, where anticipated future tax cuts boost permanent income and consumption, and the credibility effect, where lower sovereign risk premiums reduce interest rates and "crowd in" private investment. A credible consolidation serves to signal a "break from the past", which can bolster market confidence and stabilize asset prices. Beyond demand-side impacts, the supply-side or labor market channel emphasizes that the composition of adjustment is a critical determinant of success. Reducing government wages and transfers is often viewed as more effective for competitiveness and growth than increasing distortionary taxes, which can negatively impact labor supply. Alesina et al. (2019) attribute this asymmetry to expectations. Expenditure-based adjustments signal a lower future path of public spending, and therefore a reduced future tax burden on households and firms, while tax-based adjustments that leave entitlement spending unaddressed generate the opposite expectation. Despite these potential benefits, Alesina et al. (2019) acknowledge that short-term adjustment costs to output, consumption, and welfare are frequently unavoidable and carry significant distributional consequences. In the long term, however, fiscal consolidation is generally viewed as beneficial for growth, provided the resulting fiscal space is utilized to reduce distortionary taxation (Alesina et al., 2019).

Empirically, fiscal consolidations have been shown to, on average, increase inequality. The evidence is most consistent for advanced OECD economies (Ball et al., 2013; Fabrizio and Flamini, 2015; Heimberger, 2020), and is robust to the measurement of inequality and to the identification strategy (Furceri et al., 2015).[5] The size of the response is contested. Woo et al.

[5]For an EU-specific analysis, see Heimberger and Matzner (2026).

(2013) and Heimberger (2020) find larger inequality effects for large adjustments in terms of GDP, while Agnello and Sousa (2014) argue the opposite, and Konstantinou (2024), using dose-response functions, finds evidence for both small and large adjustments.[6] Longer episodes tend to be more regressive (Woo et al., 2013; Heimberger, 2020), as do consolidations undertaken following a financial crisis or during low-growth periods (Agnello and Sousa, 2014; Heimberger, 2020). For LAC the evidence is more recent but qualitatively consistent, with Komatsuzaki (2024) and de Souza Cardoso and Barbosa de Carvalho (2023) documenting a positive inequality response to consolidation episodes.

The composition of the adjustment is not trivial. On average, expenditure-based consolidations are less costly in output terms than those based on taxes (Alesina et al., 2019; Ando et al., 2025), but they tend to produce larger increases in inequality in OECD economies (Woo et al., 2013; Ball et al., 2013; Fabrizio and Flamini, 2015; Heimberger, 2020). The pattern is less settled in LAC. Komatsuzaki (2024) shows that the inequality response is larger when the adjustment relies primarily on tax hikes, with expenditure-based austerity protecting social security spending and falling instead on capital and goods-and-services lines, and with a larger effect for non-commodity exporters. de Souza Cardoso and Barbosa de Carvalho (2023), however, find for the case of South America an increase in the Gini index for spending-based consolidations but no significant effect for tax-based episodes, with their results robust to alternative controls, lag structures, sample compositions, and to the use of both narrative and CAPB-based identification, leaving the compositional pattern in the region an open question that Section 5 revisits by letting state capacity condition the response.

A complementary literature, more recent and directly related to the heterogeneity dimension exploited in this thesis, has begun to explore how state capacity and institutional quality condition the macroeconomic response to fiscal consolidations. The origins of this state capacity are themselves an object of study. Besley and Persson (2009) model fiscal and legal capacity as complementary, historically path-dependent investments shaped by political institutions and the threat of conflict, providing a structural rationale for treating QoG as a proxy for accumulated state capacity rather than a purely contemporaneous institutional label. QoG has also been linked directly to distributive outcomes outside the fiscal-consolidation setting studied here. Rothstein et al. (2012) find that it predicts the size and generosity of the welfare state across a panel of OECD countries, in line with the transfer- and spending-reach channel developed in Section 3, while Gupta et al. (2002) find that corruption raises income inequality by biasing the tax system toward those with political access and by degrading the targeting of social spending, echoing the tax-progressivity channel developed there. This is part of the case for organizing the empirical analysis around QoG rather than region or informality alone. Much of the existing evidence documents this indirectly through the lens of labor informality, which this thesis treats as an observable symptom of weak state capacity (a narrow, poorly enforced tax base and thin transfer coverage) rather than as an independent structural feature in its own right. Using local projections on a panel of more than 140 countries, Colombo et al. (2024) document that the public expenditure multiplier declines sharply with the size of the informal sector, a result they rationalize through a two-sector New-Keynesian model in which fiscal shocks shift demand toward the untaxed sector. The result is sharpened in two recent papers by Bechchani (2025a) and Bechchani (2025b), both based on the narrative dataset of Carrière-Swallow et al. (2021) for 14 LAC countries from 1989 to 2016. The first documents that the contraction in real GDP following a consolidation is concentrated in low-informality settings, with output declining by 1.12% on impact after tax hikes and by 2.77% after spending cuts, while high-informality economies display a substantially muted response. The second

[6] Small adjustments are those below 0.20% of GDP, large ones those above 2.17%.

isolates the mechanism. In high-informality economies, the twin-deficit channel operates as an external buffer, with real exchange rate depreciation and current-account adjustment absorbing part of the shock. More directly on the governance side, Jalles et al. (2025) show for a panel of emerging-market and developing economies, with a focus on LAC, that higher informality and lower corruption raise the likelihood of a consolidation being undertaken in the first place, linking institutional quality to the incidence of austerity itself and not only to its transmission. Together, this literature establishes a robust macroeconomic prior that state capacity (observed through informality, corruption, or both) modulates the transmission of fiscal consolidation, and motivates the question of whether the same modulation operates on the distributional margin.

A second relevant body of work approaches the distributional question from a static, accounting perspective via the analysis of fiscal incidence. The Commitment to Equity (CEQ) framework, developed in Lustig (2022) and applied to sixteen Latin American economies in Lustig (2017), decomposes the redistributive impact of the fiscal system into the contributions of direct taxes, indirect taxes, cash transfers, in-kind transfers, and subsidies. Two findings are central to the present analysis. First, redistribution in LAC operates overwhelmingly through the expenditure side, in particular through in-kind transfers in education and health; direct taxes contribute relatively little because the income tax base is narrow. Second, in several LAC countries the burden of consumption taxes is high enough that some segments of the poor are net contributors to the fiscal system before in-kind transfers are accounted for, underscoring that social expenditure is effectively the only redistributive channel of consequence for the bottom of the distribution. Goñi et al. (2011) document an earlier and broader version of the same point. A complementary line nuances the standard presumption that indirect taxation is uniformly regressive. Bachas et al. (2024), using household expenditure microdata for 32 countries, show that consumption taxes are de facto progressive in developing economies because the budget share spent in informal stores declines sharply with income, so that effective tax rates on a broad consumption tax rise with income. The implication for the present analysis is that the marginal incidence of a tax-based consolidation in a high-informality economy cannot be inferred mechanically from the nominal regressivity of the VAT; the empirical analysis must let the data adjudicate.

Taken together, the literature establishes three relatively well-developed but disconnected bodies of evidence: a literature on the average distributional effect of consolidations (Ball et al., 2013; Woo et al., 2013; Fabrizio and Flamini, 2015; Heimberger, 2020; Komatsuzaki, 2024; de Souza Cardoso and Barbosa de Carvalho, 2023); a literature on the macroeconomic role of state capacity, observed through informality and corruption, in shaping the aggregate response to fiscal consolidation (Colombo et al., 2024; Bechchani, 2025a,b; Jalles et al., 2025); and a literature on the static fiscal incidence of taxes and transfers in LAC (Lustig, 2017; Goñi et al., 2011; Bachas et al., 2024). None of these literatures directly answers the question this thesis poses. The first does not exploit institutional quality as a structural source of heterogeneity in the distributional response. The second focuses on output, not on distribution. The third is accounting-based and does not speak to the dynamic response of inequality to a fiscal shock. This thesis sits at the intersection of the three. It studies the dynamic distributional response to consolidation episodes, lets that response vary continuously with QoG rather than partitioning by region, and applies a consistent local projections framework across OECD and LAC, identified through the narrative approach. In doing so, it provides, to the best of the author's knowledge, the first systematic test of whether the macroeconomic modulation of fiscal consolidation by state capacity, documented through its informality channel by Bechchani (2025a) and Colombo et al. (2024), extends to the distributional margin.

Accounting for the aforementioned literature, this thesis seeks to answer two main ques-

tions.

Given that the literature has already established that fiscal consolidations are associated with an increase in the Gini coefficient, the first and main question does not attempt to re-establish this fact but rather asks whether these austerity episodes have a heterogeneous effect depending on the country's QoG. Countries with higher QoG have historically built broader tax bases, deeper transfer coverage, and stronger automatic stabilizers, so that the same fiscal adjustment may translate into a quantitatively different distributional response than in lower-QoG settings. Second, as tax-based and expenditure-based consolidations could have different distributional effects, and given that the underlying tax and transfer structures interact tightly with QoG, this thesis seeks to answer whether the compositional patterns documented separately for OECD and LAC survive or reverse when the same identification framework is applied to a pooled sample in which QoG is allowed to condition the response.

Underlying both questions is a robustness concern, namely whether the answers hold across the battery of checks detailed in Appendix A, spanning alternative measures and cutoffs of QoG, sample composition, and potential confounders.

Two features of the design that follows are new to the literature just reviewed. No existing study of the distributional effects of consolidation organizes a pooled OECD and LAC sample around a structural measure of state capacity rather than around region or income level. Nor does any estimate the response of redistribution itself, as opposed to the response of Gini levels, which is where the effect documented below turns out to be concentrated.

3 Conceptual framework

For the theoretical foundations of this analysis, this thesis adapts a simplified version of the heterogeneous-household model in Ferriere and Navarro (2025), abstracting from their New Keynesian block, monetary policy rule, and discount-factor heterogeneity to isolate a stationary incomplete-markets environment suited to the fiscal channels studied here. Assume time to be discrete and indexed by $t = 0, 1, 2, \ldots$. The economy consists of a continuum of households, a representative firm, and a government. The firm operates a constant-returns-to-scale production technology in capital and labor, given by $Y = K^{\alpha} L^{1-\alpha}$, where K, L, and Y denote capital, labor, and output, respectively. Both inputs are supplied by households.

For the households, let $V(a,x)$ be the value function of a household with assets a and idiosyncratic productivity x which follows a Markov process with transition probabilities $\pi_x(x',x)$ such that:

$$V(a,x) = \max_{c,h,a'} \left\{ \log(c) - Bh + \beta \mathbb{E}_{x'} \left[V(a',x') \mid x \right] \right\}$$

$$\text{subject to} \quad \begin{aligned} c + a' &\leq wxh + (1+r)a - \mathscr{T}(wxh, ra) + T, \\ h &\in \{0, \bar{h}\}, \quad a' \geq 0 \end{aligned}$$

where c and h denote consumption and hours worked, respectively, and w and r denote wages and interest rates, respectively. Households face a distortionary tax $\mathscr{T}(wxh, ra)$, which depends on labor income wxh and capital earnings ra, and receive a lump-sum transfer T.

The representative firm chooses labor and capital demand to maximize current profits every period:

$$\Pi = \max_{K,L} \left\{ K^{\alpha} L^{1-\alpha} - wL - (r+\delta)K \right\}$$

where δ is the depreciation rate of capital. The optimality conditions for the firm are standard, equalizing marginal productivities to the cost of each factor.

Finally, the government's budget constraint is given by

$$G + (1+r)D + T = D + \int \mathscr{T}(wxh, ra)\, d\mu(a,x)$$

where D is the government's debt and $\mu(a,x)$ is the measure of households with state (a,x) in the economy. In a steady state, government spending G is kept constant.

Rather than being solved or calibrated, the model above is used throughout this section as an organizing device for the channels through which a fiscal consolidation can affect the income distribution: taxation through $\mathscr{T}(\cdot)$, the transfer T, and government spending G. The hypotheses derived below combine this structure with additional assumptions about how state fiscal capacity, proxied here by QoG, alters each of these channels: the shape and enforcement of $\mathscr{T}(\cdot)$, the coverage of T across the productivity distribution, and the size of G relative to the economy. It is these assumptions, rather than a closed-form solution or a calibrated quantitative exercise, that generate the testable predictions below.

In the case of this model, a fiscal consolidation based on an increase in $\mathscr{T}(\cdot)$ would restrict the consumption and savings ability of wealthier households more than that of hand-to-mouth individuals. Since $\mathscr{T}$ taxes both labor income wxh and capital earnings ra, and hand-to-mouth households hold negligible assets and therefore derive almost no capital income, wealthier households draw on a strictly broader tax base than hand-to-mouth households, whose taxable resources are confined to labor earnings. The same increase in $\mathscr{T}(\cdot)$ therefore extracts more from wealthier households, implying a decrease in income inequality. A cut in transfers, by contrast, falls relatively harder on low-income households, for whom transfers represent a larger share of total resources. In this framework, therefore, only expenditure-based consolidations are regressive, in the specific sense of raising inequality in disposable income. That is a statement about the level of the disposable Gini, and it does not by itself settle what happens to the distance between the market and the disposable Gini, the object measured below, which depends on how market income moves as well.

The lump-sum transfer T plays a distinct redistributive role depending on where a household sits in the productivity distribution. Since high x households accumulate more assets and earn more labor income over time, T represents a smaller share of their total resources relative to low x households, for whom it may constitute a substantial fraction of disposable income. In this sense, even a uniform transfer is implicitly progressive. The key implication is that the distributional weight of T is not symmetric across the productivity distribution. Cuts to it fall hardest on those least able to substitute through savings or labor supply adjustment, making the transfer the primary channel through which expenditure-based consolidations generate regressive outcomes in this framework. Reducing T has welfare consequences beyond the direct income loss. For households near the borrowing constraint, the transfer serves as an implicit consumption floor during low-productivity spells. Its removal forces hand-to-mouth households to adjust consumption one-for-one with the cut, while wealthier, unconstrained households can buffer the shock through their asset holdings. This asymmetry amplifies the inequality-increasing effect of transfer cuts beyond what a comparison of income levels alone would suggest.

A related channel operates through government spending G. Public expenditure on education, healthcare, and infrastructure functions as an in-kind supplement to private income, and lower-income households rely on it more heavily, both because their private resources are more limited and because their capacity to substitute toward private alternatives when public provision deteriorates is weaker. Cuts to G therefore generate a regressive incidence even when

nominally uniform, reducing the effective consumption of lower-income households by more in relative terms. This effect is most pronounced for social spending components, health and education, which serve as near-complete substitutes for private expenditure at the bottom of the distribution. This channel is an additional assumption of the kind noted at the outset of this section rather than a property of the model, in which G enters only the government's budget constraint and the goods-market clearing condition and therefore carries no distributional content of its own.

While the baseline model assumes that all households participate symmetrically in the tax system and in the transfer scheme, in practice the state's fiscal capacity, proxied here by QoG, reshapes both the tax function $\mathscr{T}(\cdot)$ and the de facto progressivity of T and G in ways that condition the distributional response to consolidation. In low-QoG economies, weak enforcement capacity leaves a sizable mass of households outside the formal tax net, an informal wedge that mechanically narrows the base on which any tax-based consolidation can operate. Where the tax authority lacks the capacity to broaden this base, governments lean disproportionately on indirect taxes, notably value-added taxes, which fall regressively on consumption and reach informal households despite their absence from the income tax rolls. The implicit progressivity of $\mathscr{T}(\cdot)$ that prevails in high-QoG settings, where the bulk of revenue comes from progressive income and corporate taxation administered over a broad formal base, is therefore eroded as QoG falls, and tax-based consolidations are expected to become most disequalizing, in the sense of raising inequality in disposable income, where the state is least able to tax progressively.[7]

That prediction concerns the level of the disposable Gini, and it does not carry over to the gap between the two Ginis, which is the object the empirical analysis measures and which Hypothesis 2 below expects to respond most strongly in the opposite, high-QoG direction. A tax increase that lands regressively raises disposable inequality without widening the redistributive wedge. Whether a consolidation widens or narrows redistribution is therefore not settled by the incidence of the adjustment alone. Redistribution is the distance between the market and the disposable Gini, so it widens whenever market inequality deteriorates by more than disposable inequality does, and the forces at work do not all pull the same way. On the spending side, the direct incidence of the adjustment removes support from the households that depend on it most, pushing disposable inequality up and closing the gap. On the tax side it runs the opposite way: a progressive increase in $\mathscr{T}(\cdot)$ extracts proportionally more from the top of the distribution, compressing disposable income and widening the gap on its own. Cutting across both, the macroeconomic contraction that follows depresses employment and earnings among lower-productivity workers more than among higher-productivity ones, pushing market inequality up and widening the gap regardless of how the adjustment was achieved.

Which force dominates depends on how much of the fiscal system survives the adjustment and keeps operating. A consolidation trims part of T and G, but in a high-capacity state the remainder is large and continues to absorb shocks: unemployment insurance still pays out, the income tax still takes progressively, and means-tested transfers still expand as earnings fall. A substantial share of the market-income deterioration is therefore intercepted before it reaches disposable income, so the disposable Gini rises by less than the market Gini and the measured wedge widens. This is how an expenditure-based consolidation can widen the gap even though its direct incidence works to close it. The cut does push disposable inequality up,

[7] This mechanism is formalized by Ayres et al. (2025), who extend a heterogeneous-agent incomplete-markets model of the same class used here with an explicit formal-informal employment margin, calibrated to Brazil, and show that the redistributive potential of statutory tax progressivity is bounded by how steeply the cost of informal employment rises with productivity, exactly the constraint that a low-QoG $\mathscr{T}(\cdot)$ is assumed to face in the framework above.

as the preceding paragraphs describe, but where a large fiscal system survives the adjustment the market side deteriorates by more, and the net movement of the gap is positive. In a low-capacity economy none of these forces is strong. A sizable informal sector sits outside the formal labor market and absorbs much of the macroeconomic shock, so market inequality moves less to begin with, and the thin surviving system has little capacity to convert whatever deterioration does occur into a wider gap. Both Ginis move little, and redistribution barely responds.

The prediction that follows is therefore not that austerity is redistributive. It is that a consolidation widens the measured gap between market and disposable inequality only where a large fiscal apparatus survives it, whether by intercepting part of the damage to market incomes before it reaches disposable income or by collecting progressively from the top of the distribution. Where that apparatus is thin, neither route is open and the gap barely moves.

Based on the conceptual framework outlined above and the empirical patterns documented in the literature, two hypotheses are proposed.

First, redistribution is expected to widen more following a fiscal consolidation in high-QoG economies than in low-QoG ones. The underlying mechanism is not that strong institutions fail to protect vulnerable households, but that QoG proxies for state fiscal capacity. Economies with higher QoG have historically built broader tax bases, deeper transfer coverage, and a correspondingly smaller informal sector, so that a nominal fiscal shock of a given size propagates through a larger, more consequential formal fiscal system and widens redistribution by more. In low-QoG economies, by contrast, a sizable informal sector sits outside the reach of both taxation and transfers and absorbs much of the shock, muting its footprint on the redistributive gap.

Second, the composition of the adjustment is expected to condition the strength, though not the direction, of this state-capacity effect. Both expenditure-based and tax-based consolidations should widen redistribution more in high-QoG than in low-QoG economies, but through different channels: expenditure-based consolidations operate through the size and reach of the transfer and spending system a high-capacity state can sustain, while tax-based consolidations operate through the progressivity and enforcement capacity of a high-capacity tax administration. Because the tax channel acts more directly on the redistributive gap, compressing disposable income at the point of collection rather than through a slower, indirect channel operating via market income, tax-based consolidations are expected to produce a larger and more persistent redistributive effect in high-QoG economies than expenditure-based ones.

In the notation of the empirical strategy and results below, Hypothesis 1 corresponds to a positive γ^h in equation 3, and Hypothesis 2 to a positive θ^h in equations 4 and 5.

4 Empirical strategy

4.1 Data

Seven different data sources are utilized for 30 countries from 1996 to 2019. For inequality, specifically disposable Gini, market Gini, and redistribution, the Standardized World Income Inequality Database (SWIID)[8] is used, as presented by Solt (2020). A narrative fiscal consolidation database by Adler et al. (2024)[9] is used to identify consolidation episodes. For government expenditure by function and revenue by tax type, a database by Gethin (2023) is used, which also provides GDP and population. In terms of control variables, real GDP is taken

[8] Version 9.9.

[9] The data used are those reported in the paper itself.

from the Penn World Table[10] (Feenstra et al., 2015), and other variables, such as inflation, trade openness, unemployment, and urban population, are sourced from the World Development Indicators, with government debt taken from the fiscal-space database of Kose et al. (2022). Data on QoG, used to construct the heterogeneity dimension, are taken from the Worldwide Governance Indicators (WGI) (World Bank, 2025), averaging the estimates of control of corruption, government effectiveness, rule of law, and regulatory quality; the continuous interaction is evaluated at the 25th and 75th percentiles of this index, with a median-split and a first-/last-decile version reported as robustness checks in Appendix A.1.

Each of the four components entering QoG is an estimate from the WGI's unobserved-components model, which aggregates perceptions drawn from expert assessments and from firm and household surveys into an index expressed in units of a standard normal distribution (World Bank, 2025). Control of corruption captures perceptions of the exercise of public power for private gain; government effectiveness, the quality of public services and of policy formulation and implementation; rule of law, confidence in contract enforcement, property rights, and the courts; and regulatory quality, the government's ability to design and enforce sound regulation. The four components are highly correlated in the sample, which motivates the equal-weight average used as the baseline; Appendix A.2 shows the results are unchanged under a data-driven weighting, and Appendix A.9 examines each component on its own.

Regarding the inequality data, SWIID is chosen over alternative databases on coverage grounds. To the best of the author's knowledge, it is the only source providing annual, methodologically standardized series for both the market and the disposable Gini across all 30 OECD and LAC countries in the sample over 1996–2019. The Luxembourg Income Study, on which SWIID's standardization is anchored, offers rawer microdata but sparse and irregular coverage of LAC; the World Inequality Database focuses on pretax top income shares rather than the market and disposable pair that the redistribution measures used here require. This coverage has a cost. SWIID is a model-based imputation, and its standardization has been criticized for the difficulty of tracing how much of any given series reflects observation rather than model (Jenkins, 2015). Two consequences for this analysis follow: the imputation uncertainty is not propagated into the standard errors reported below, and year-to-year movements in the Gini series are smoother than in the underlying surveys, which works against finding sharp dynamic responses and makes the estimates conservative in that specific sense.

Fiscal consolidations have historically been identified through one of two approaches: the "narrative" approach (Romer and Romer, 2010; Devries et al., 2011) and the CAPB or statistical approach (Alesina and Ardagna, 2010). The first identifies episodes from explicit government intentions, as stated in budget documents and official statements. This "policy action" approach benefits from insulating the identified fiscal measures from the business cycle, owing to their "bottom-up" construction, which in turn reduces identification problems and lowers the risk of reverse causality. Nevertheless, this approach is not free of drawbacks as it depends on the subjective judgment of those who build the database and does not eliminate endogeneity (Jordà and Taylor, 2016). Replicating it for a dynamic panel is also logistically demanding.

The second approach identifies fiscal consolidations through an ad-hoc change in the CAPB, a measure that adjusts the budget balance for the influence of the business cycle on government revenues and spending; there is no generalized consensus on how large that change must be. For example, this approach would prevent a normal cyclical downturn in government revenues from being misclassified as an episode of fiscal expansion. It benefits from being more easily replicable than the narrative approach, with CAPB data publicly available from international organizations or computed using a trend filter of choice. Moreover, as there is no clear consensus

[10] Version 11.

on the optimal method for estimating potential output, one must remain aware of the shortcomings of applying filtering techniques (Jalles et al., 2025). Furthermore, endogeneity problems arise because of the inherent link between these indicators and economic cycles. Even if variations in the CAPB did provide an accurate measure of discretionary fiscal policy changes, the underlying drivers of these shifts could still be tied to reactions to cyclical fluctuations. This is because cyclical adjustment techniques do not fully strip out the influence of large swings in economic activity and asset prices from fiscal indicators, leading to movements in the CAPB that are not necessarily attributable to fiscal policy itself (Devries et al., 2011).

Aware of the shortcomings and advantages of each approach, this thesis adopts the narrative approach as its identification strategy, given the endogeneity and filtering concerns discussed above for the CAPB alternative. An up-to-date database by Adler et al. (2024) is used, which follows previous fiscal consolidation databases such as Devries et al. (2011), Alesina et al. (2018), and David and Leigh (2018). This database is compiled by analyzing policymakers' stated intentions and subsequent actions in contemporaneous policy documents[11], and by isolating those measures whose primary motivation is deficit reduction. Because these fiscal measures are reactions to past decisions and economic conditions, rather than to anticipated future developments, they are unlikely to be systematically linked with other short-term factors influencing output and can therefore be legitimately used to estimate the macroeconomic effects of fiscal consolidations (Romer and Romer, 2010). These episodes are identified for 31 countries, 17 of which are from OECD from 1978 to 2020, and 14 from LAC[12] from 1989 to 2020. [13] Table 8 in Appendix B.2 lists these episodes restricted to the 1996–2019 analysis sample actually used in the estimations below (and excluding Bolivia), rather than this full raw history. Within that sample the database records 191 consolidation episode-years across the 30 countries, out of 720 country-year observations in total. Besides including the country and year of the fiscal consolidation, the database also contains information regarding the expenditure cuts and tax hikes derived from the episode in terms of GDP. For the narrative approach, a fiscal consolidation is defined as expenditure-based (tax-based) if the reduction of the government's spending is greater (lower) than the increase in revenue in terms of GDP.

Table 1 reports descriptive statistics, mean, standard deviation, minimum, maximum, and number of observations, for every variable used in the empirical analysis: the four outcome variables, the fiscal consolidation size, QoG and its four components, and the six macroeconomic controls, all computed over the analysis sample.[14]

[11]These include: Budget Reports, central bank reports, Convergence Programmes and Stability Programmes submitted by the authorities to the European Commission, as well as IMF reports and OECD Economic Surveys.

[12]OECD: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Ireland, Italy, Japan, Netherlands, Portugal, Spain, Sweden, United Kingdom and United States. LAC: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, Ecuador, Guatemala, Jamaica, Mexico, Paraguay, Peru and Uruguay.

[13]For this analysis, Bolivia is excluded because, as in Komatsuzaki (2024), two of its three fiscal consolidation episodes took place during the commodity boom of 2004 and 2005.

[14]Table 10 in Appendix B.3 reports the exact definition and original data source of each of these variables.

Table 1: Descriptive statistics, analysis sample (1996–2019)

	Variable	Mean	SD	Min	Max	N
Outcomes	Market Gini	49.414	3.610	39.900	61.600	720
	Disposable Gini	37.017	9.085	22.200	56.400	720
	Absolute redistribution	12.398	8.299	−1.000	25.800	720
	Relative redistribution	25.206	16.978	−1.992	49.446	720
Fiscal consolidation	Total adjustment (% GDP)	0.263	0.683	−0.900	5.230	720
Quality of government	QoG (average)	0.824	0.962	−1.208	2.213	720
	Control of corruption	0.847	1.076	−1.603	2.484	720
	Government effectiveness	0.840	0.976	−1.372	2.322	720
	Rule of law	0.750	1.014	−1.166	2.109	720
	Regulatory quality	0.858	0.851	−1.299	2.176	720
Controls	Log real GDP per capita (millions of 2017 PPP USD)	−3.647	0.731	−5.263	−2.139	720
	Unemployment (% labor force)	7.609	3.658	1.902	26.094	695
	Inflation (% YoY)	3.922	6.636	−4.448	96.096	644
	Trade openness (% GDP)	68.499	35.753	15.636	249.746	720
	Government debt (% GDP)	62.904	38.833	3.901	236.381	696
	Urban population (% total)	75.699	12.632	43.989	98.041	720

Source: Own elaboration.

4.2 Methodology

As is standard in the literature, this thesis follows Local Projections (LP) methods (Jordà, 2005; Jordà and Taylor, 2025). For a discussion of Local Projections, their properties, and their comparison to VARs, see Appendix B.1. Let $\Delta Y_{i,t+h} = Y_{i,t+h} - Y_{i,t-1}$ be the cumulative difference h periods forward. Throughout, $Y_{i,t}$ denotes one of four SWIID outcome variables (Solt, 2020), all used in levels: the market Gini, $\text{gini_mkt}_{i,t}$, the Gini index of inequality in equivalized (square-root-scale) household market, pre-tax and pre-transfer, income; the disposable Gini, $\text{gini_disp}_{i,t}$, the Gini index of inequality in equivalized household disposable, post-tax and post-transfer, income; and absolute and relative redistribution, $\text{abs_red}_{i,t}$ and $\text{rel_red}_{i,t}$, the point and percentage reduction, respectively, in market-income inequality due to taxes and transfers, defined as:

$$\text{abs_red}_{i,t} = \text{gini_mkt}_{i,t} - \text{gini_disp}_{i,t}, \quad \text{rel_red}_{i,t} = \frac{\text{gini_mkt}_{i,t} - \text{gini_disp}_{i,t}}{\text{gini_mkt}_{i,t}} \times 100$$

None of the four is logged, so $\Delta Y_{i,t+h}$ is a cumulative point change rather than a percentage change: Gini points for the market Gini, the disposable Gini, and absolute redistribution, and percentage points of the redistribution ratio for relative redistribution. This is reported throughout simply as "cumulative difference".

The empirical strategy builds up in four steps, moving from the pooled average effect toward increasingly granular conditioning, in the same order as the results in Section 5.

The first step establishes the average, unconditional effect of a fiscal consolidation:

$$\Delta Y_{i,t+h} = \beta^h \mathbf{FC}_{i,t} + \phi^h X_{i,t} + \alpha_i^h + \varepsilon_{i,t+h} \tag{1}$$

This is estimated by OLS for each horizon $h \in \{-2, \ldots, 5\}$, separately for each of the four outcome variables, with Driscoll–Kraay standard errors (Driscoll and Kraay, 1998). Here $\mathbf{FC}_{i,t}$, the fiscal consolidation (FC) size, is the net size of the fiscal adjustment, in percent of GDP, continuous and equal to zero in country-years without a consolidation episode, and X is a vector

of controls comprising two lags of the first difference of the dependent variable,[15] the lag of the log of GDP per capita, and the lags of the unemployment rate, of inflation, of trade openness (exports plus imports as a share of GDP), of government debt (as a share of GDP), and of the urban population share. In the interacted specifications below, X also absorbs the constituent level terms of each interaction: $\mathbf{QoG}_{i,t}$, the relevant composition dummy, and, in equations 4 and 5, their product. α_i^h is a country fixed effect, while $\varepsilon_{i,t+h}$ is the error component. β^h is the marginal cumulative response h periods after a one-percentage-point-of-GDP increase in the size of a fiscal consolidation, pooling all countries regardless of their QoG or the composition of the adjustment.

The second step asks whether the composition of the adjustment, tax-based versus expenditure-based, matters on its own, still without conditioning on QoG. Let $\mathbf{D}_{i,t}^G = \mathbb{1}(\Delta G_{i,t} > \Delta T_{i,t})$ denote an expenditure-based consolidation and $\mathbf{D}_{i,t}^T = \mathbb{1}(\Delta T_{i,t} > \Delta G_{i,t})$ a tax-based one, where $\Delta T_{i,t}$ is the revenue increase and $\Delta G_{i,t}$ the expenditure reduction attributed to the episode, both in percent of GDP and signed so that a negative value records a component moving against the consolidation:

$$\Delta Y_{i,t+h} = \beta^h \mathbf{FC}_{i,t} + \lambda^h \mathbf{FC}_{i,t} \times \mathbf{D}_{i,t} + \phi^h X_{i,t} + \alpha_i^h + \varepsilon_{i,t+h} \tag{2}$$

estimated separately with $\mathbf{D}_{i,t} = \mathbf{D}_{i,t}^G$ and with $\mathbf{D}_{i,t} = \mathbf{D}_{i,t}^T$. Both indicators use strict inequalities, so the four episodes in which the two components are exactly equal fall into neither group: of the 191 episodes, 114 are expenditure-based, 73 tax-based, and 4 evenly split. This is why each arm is estimated separately rather than as a single dummy and its complement, which would assign those episodes to one type by construction. Here λ^h is the differential marginal effect, per percentage point of GDP, of the relevant composition type relative to the rest of the sample, still pooling across all levels of QoG.

The third step introduces QoG as a moderator, testing Hypothesis 1:

$$\Delta Y_{i,t+h} = \beta^h \mathbf{FC}_{i,t} + \gamma^h \mathbf{FC}_{i,t} \times \mathbf{QoG}_{i,t} + \phi^h X_{i,t} + \alpha_i^h + \varepsilon_{i,t+h} \tag{3}$$

where $\mathbf{QoG}_{i,t}$ is the country's QoG in year t, as defined in Section 4.1. γ^h is the marginal effect of QoG on the cumulative response, so that $\beta^h + \gamma^h \times \mathbf{QoG}_{i,t}$ gives the marginal response, per percentage point of GDP of consolidation size, at any given level of QoG; throughout the results, this combination is evaluated at the 25th and 75th percentiles of the sample QoG distribution, labeled Low QoG and High QoG, respectively.[16] A positive γ^h indicates that the effect of a consolidation grows with QoG; the hypotheses anticipate $\gamma^h > 0$ for redistribution. For readability, equations 1 through 5 display only the contemporaneous consolidation term, but each consolidation variable and its interactions also enter with two lags;[17] all reported coefficients are the contemporaneous ones.

The final step, testing Hypothesis 2, combines both dimensions at once, estimating the QoG-

[15] Under the cumulative transformation the lagged dependent variable enters in first differences, $\Delta Y_{i,t-1}$ and $\Delta Y_{i,t-2}$, and at non-positive horizons these lags are shifted back by a further $|h|$ periods. At $h = -2$ they are therefore $\Delta Y_{i,t-3}$ and $\Delta Y_{i,t-4}$, which do not overlap with the left-hand side $Y_{i,t-2} - Y_{i,t-1}$, so the pre-shock coefficient reported in Appendix A.4 is not mechanical.

[16] An alternative moderator would be the country's own pre-consolidation level of redistribution. QoG is preferred on two grounds: a moderator constructed from the lagged dependent variable would confound genuine cross-country heterogeneity with mean reversion in the outcome, and QoG is a slow-moving characteristic with direct theoretical content in Section 3, entered contemporaneously in the baseline and lagged one period in Appendix A.3.

[17] The lagged consolidation terms are included as additional regressors and omitted from the displayed equations only for clarity. All coefficients reported throughout correspond to the contemporaneous terms.

conditional response separately for expenditure-based consolidations (equation 4) and tax-based consolidations (equation 5):

$$\begin{aligned}\Delta Y_{i,t+h} =& \beta^h \mathbf{FC}_{i,t} + \gamma^h \mathbf{FC}_{i,t} \times \mathbf{QoG}_{i,t} + \lambda^h \mathbf{FC}_{i,t} \times \mathbf{D}^G_{i,t} + \theta^h \mathbf{FC}_{i,t} \times \mathbf{QoG}_{i,t} \times \mathbf{D}^G_{i,t} \\ &+ \phi^h X_{i,t} + \alpha^h_i + \varepsilon_{i,t+h}\end{aligned} \tag{4}$$

$$\begin{aligned}\Delta Y_{i,t+h} =& \beta^h \mathbf{FC}_{i,t} + \gamma^h \mathbf{FC}_{i,t} \times \mathbf{QoG}_{i,t} + \lambda^h \mathbf{FC}_{i,t} \times \mathbf{D}^T_{i,t} + \theta^h \mathbf{FC}_{i,t} \times \mathbf{QoG}_{i,t} \times \mathbf{D}^T_{i,t} \\ &+ \phi^h X_{i,t} + \alpha^h_i + \varepsilon_{i,t+h}\end{aligned} \tag{5}$$

In both, λ^h is the differential marginal response when the relevant composition dummy equals one, and θ^h is the triple-interaction term measuring how that compositional differential itself varies with QoG. The resulting marginal cumulative response, evaluated at a given QoG level q, is $\beta^h + \gamma^h q$ absent the relevant composition and $\beta^h + \gamma^h q + \lambda^h + \theta^h q$ under it, evaluated as elsewhere at the 25th and 75th percentiles of QoG.

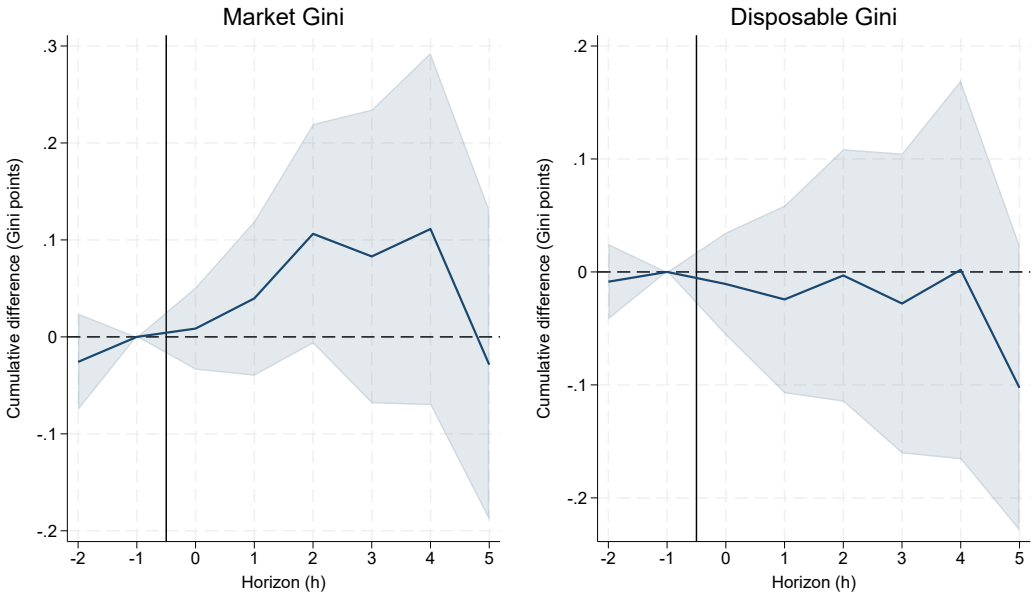
5 Results

Results follow the four-step sequence set out in Section 4, each step conditioning on one further dimension than the last. The first subsection reports the average, unconditional effect of a fiscal consolidation on the four outcome variables. The second estimates that effect separately by composition of the adjustment, tax-based versus expenditure-based, still without conditioning on QoG. The third introduces QoG as a moderator of the aggregate response. The fourth interacts both dimensions, estimating the QoG-conditional response separately for each type of consolidation.

5.1 Average effect

This subsection establishes the average, unconditional effect of a fiscal consolidation on the four outcome variables, before conditioning on QoG. This is the benchmark against which the literature's standard finding that austerity is disequalizing can be compared (Ball et al., 2013; Woo et al., 2013; Fabrizio and Flamini, 2015; Heimberger, 2020; Komatsuzaki, 2024; de Souza Cardoso and Barbosa de Carvalho, 2023). If fiscal consolidations were disequalizing in the same way and to the same degree everywhere, this average effect would already tell most of the story that matters. It does not. The pooled, unconditional picture turns out to be a poor guide to what is happening underneath it.

Figure 3: Average consolidation effect on inequality (unconditional on QoG)

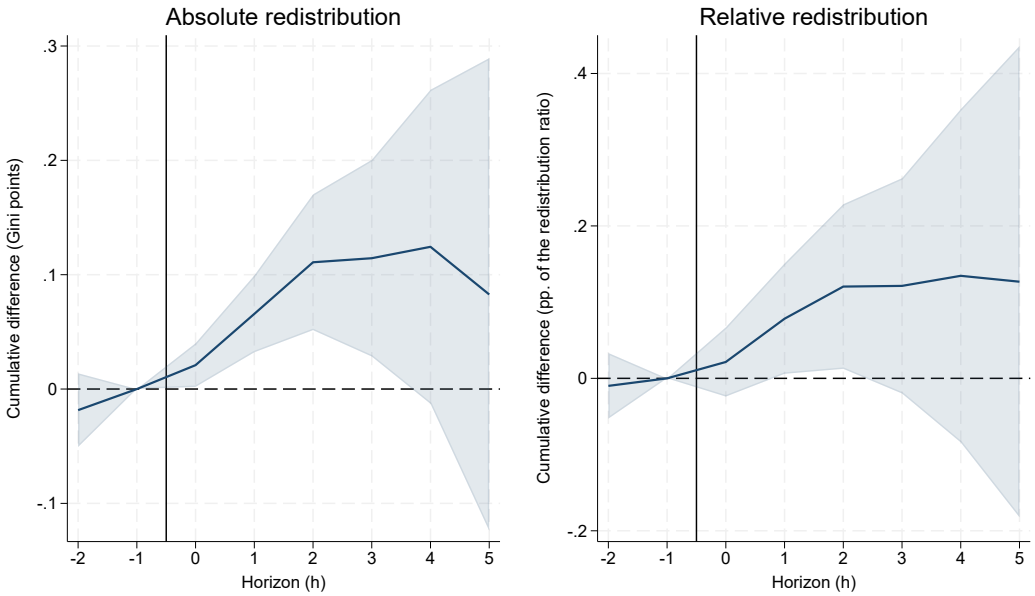


Note: Impulse response function of a fiscal consolidation on the market and disposable Gini as cumulative difference, without conditioning on QoG. Shaded area represents a 90% confidence interval. Source: Own elaboration.

Pooling every consolidation episode in the sample together, neither Gini moves in any way that can be distinguished from noise. The 90% confidence interval for both the market and the disposable Gini includes zero at every horizon over the five years following a consolidation (Figure 3). Taken at face value, this would suggest that fiscal consolidations are, on average, distributionally neutral, a conclusion that sits uneasily with the broader literature cited above. The rest of this section argues it is instead an artifact of pooling together countries and consolidation types whose effects work in different directions and through different channels, not evidence that no such effects exist.

Redistribution offers the first hint that this null result is misleading rather than definitive (Figure 4). Both the absolute and the relative measure rise steadily over the first several post-shock years, and, unlike either Gini, their confidence intervals exclude zero over the early horizons. Absolute redistribution is significant from the year of the shock through $h = 3$, and the relative measure over $h = 1$ to $h = 2$, before both intervals widen again as the horizon lengthens. The effect is modest and short-lived, but it is there, and it is there only for redistribution. A fiscal consolidation's distributional consequences operate through the gap between the two Ginis and not through the level of either one. The remainder of the analysis, including every robustness check in Appendix A, is accordingly built around absolute and relative redistribution rather than around the Gini coefficients that motivated the question in the first place.

Figure 4: Average consolidation effect on redistribution (unconditional on QoG)



Note: Impulse response function of a fiscal consolidation on redistribution (absolute and relative) as cumulative difference, without conditioning on QoG. Shaded area represents a 90% confidence interval. Source: Own elaboration.

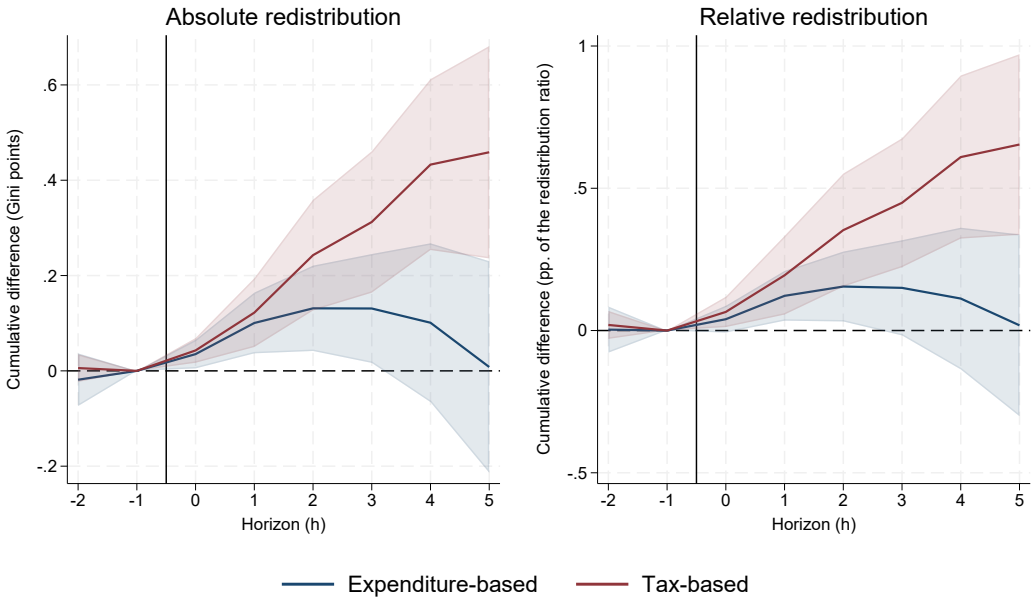
This thesis treats redistribution as the primary outcome throughout, the most direct point of contact with a fiscal consolidation's effect on the income distribution. A fiscal consolidation is by construction a change in the wedge between market and disposable income. The model of Section 3 operates on the tax function and the transfer, which jointly define that wedge, and makes no unambiguous prediction for the level of either Gini taken separately. Reporting both Gini series nonetheless serves two purposes: it locates the estimates relative to the existing literature, which has focused on levels, and it would in principle allow decomposing a widening redistributive gap into deteriorating market incomes, improving disposable incomes, or both; in the pooled estimates (Figure 3), neither margin moves significantly on its own.

5.2 Composition of adjustment

This subsection asks whether the composition of the adjustment (achieved mainly through higher taxes, or mainly through lower spending) matters on average, replicating the literature's finding that expenditure- and tax-based consolidations have different distributional consequences (Woo et al., 2013; Ball et al., 2013; Fabrizio and Flamini, 2015; Heimberger, 2020; Komatsuzaki, 2024).

The two types of consolidation behave very differently even before QoG enters the picture (Figure 5). Both raise redistribution over the first post-shock years, but they diverge sharply in magnitude and in persistence. The tax-based response is significant at every horizon and grows steadily, reaching roughly 0.31 points of absolute redistribution per percentage point of GDP by $h = 3$ and 0.46 by $h = 5$. The expenditure-based response is significant early on, peaks at about 0.13 points around $h = 2$ to $h = 3$, and fades back to zero by the end of the window. Table 2 reports the corresponding coefficients at every horizon, with each arm estimated separately as in equation 2.

Figure 5: Average consolidation effect by composition (unconditional on QoG)



Note: Impulse response function of a fiscal consolidation on redistribution (absolute and relative) as cumulative difference, without conditioning on QoG, estimated separately for each composition type as in equation 2. Tax-based consolidations are those where the increase in government revenue is greater than the decrease in government spending caused by government austerity, and expenditure-based those where the decrease in spending is the greater of the two; each line reports the response for episodes of the corresponding type. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Table 2: Consolidation effect by composition of the adjustment, by horizon (unconditional on QoG)

Outcome	Composition	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$
Absolute redistribution	Expenditure-based	0.035*	0.100***	0.131**	0.131*	0.101	0.008
		(0.018)	(0.039)	(0.054)	(0.069)	(0.101)	(0.135)
	Tax-based	0.043***	0.122***	0.243***	0.312***	0.433***	0.459***
		(0.016)	(0.044)	(0.071)	(0.090)	(0.109)	(0.136)
Relative redistribution	Expenditure-based	0.040	0.122**	0.155**	0.150	0.112	0.018
		(0.029)	(0.053)	(0.075)	(0.102)	(0.151)	(0.194)
	Tax-based	0.065**	0.194**	0.353***	0.449***	0.610***	0.654***
		(0.032)	(0.084)	(0.121)	(0.138)	(0.174)	(0.193)

Note: marginal cumulative response to a one-percentage-point-of-GDP fiscal consolidation of the corresponding composition type, $\beta^h + \lambda^h$, evaluated at each horizon h, from equation 2 estimated separately with $\mathbf{D} = \mathbf{D}^G$ and with $\mathbf{D} = \mathbf{D}^T$. Standard errors in parentheses, backed out from the 90% confidence interval under a normal approximation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Own elaboration.

This asymmetry has a straightforward incidence-based explanation. A tax increase falls directly on the income of the households that pay it, and does so progressively whenever effective tax rates rise with income, so it mechanically compresses the disposable-income distribution at the very point of collection, with no intermediate step for the redistributive effect to get lost in along the way. The measure also stays on the books, so its effect accumulates for as long as it remains in force. An expenditure cut, by contrast, reaches disposable income only indirectly, through whichever transfer or public service happens to be reduced. A reduction in an

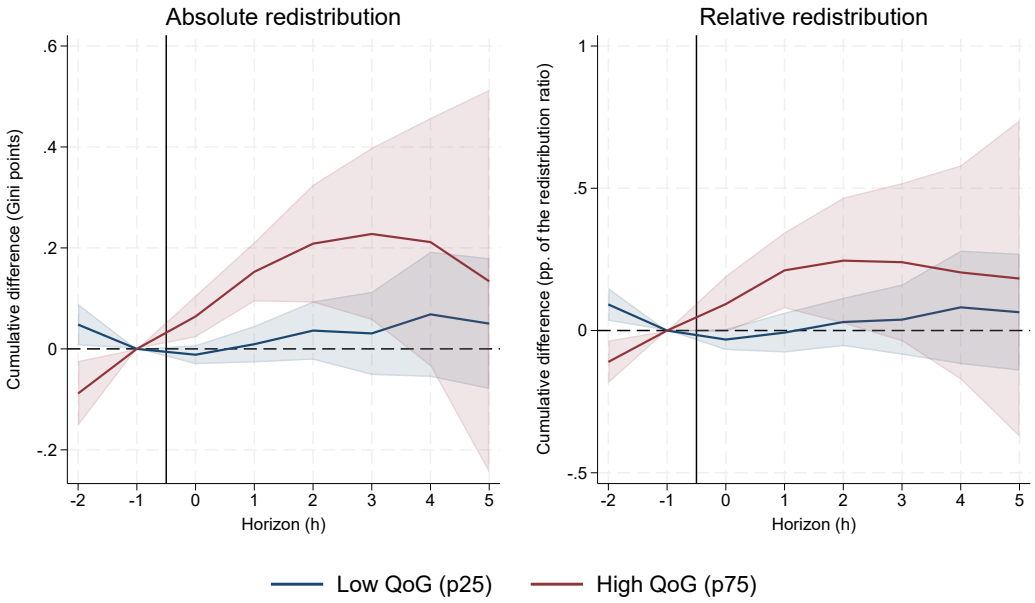
untargeted subsidy, a public-sector wage freeze, and a cut to a means-tested transfer can have very different, even opposing, distributional consequences. Pooled across thirty countries with very different public-spending systems, enough of that incidence survives to leave a measurable effect in the first years, but not one that holds as the episodes mature and individual cuts are reversed or absorbed. This ranking does not contradict the OECD evidence reviewed in Section 2, where expenditure-based adjustments are the ones that produce the larger increases in inequality (Woo et al., 2013; Ball et al., 2013). That literature measures the level of the disposable Gini, whereas the object here is the distance between the market and the disposable Gini, and a disposable Gini that rises is entirely compatible with a wedge that widens, provided market inequality rises by more. An expenditure-based consolidation can therefore be the more disequalizing of the two in levels, as that literature finds, and still be the weaker of the two at widening the wedge, as reported here, because the tax channel reaches the wedge by the more direct route of compressing the disposable distribution at the point of collection. The distributional consequences of a consolidation therefore depend on its composition and not merely on its size, a theme that becomes central once QoG is allowed to condition the response in the final part of this section.

5.3 QoG-conditional effect

Conditioning on QoG is where the central result of this thesis emerges. Figure 6 evaluates the same continuous consolidation $\times$ QoG interaction at the 25th and 75th percentiles of the governance index. The weak, narrow-window average response documented in Figure 4 separates cleanly into two very different stories, consistent with Hypothesis 1. High-QoG countries show a significant increase in absolute redistribution from the year of the shock through $h = 3$, and in relative redistribution over the narrower $h = 1$ to $h = 2$ window, while low-QoG countries show no significant response at any horizon of the estimation window.

Table 3 reports the corresponding coefficients at every horizon. In economic terms, the estimate at $h = 3$ implies that three years after a consolidation of one percent of GDP, the gap between the market and the disposable Gini is roughly 0.23 points wider in a high-QoG economy than it would have been without the shock. Against a sample mean of 12.4 points of absolute redistribution, with a standard deviation of 8.3 (Table 1), this amounts to about two percent of the average redistributive wedge per percentage point of consolidation. The response is moderate, though far from negligible for the larger adjustments in the sample, several of which exceed three percent of GDP in a single year. This early-to-medium-run divergence is robust to excluding the 2008–2009 financial crisis (Appendix A.5), and the absolute-redistribution result also holds under country- or region-specific secular trend controls; the relative-redistribution result is more sensitive to these controls, particularly the country-specific one, and both measures lose precision at the longest horizons considered, as detailed in Appendix A.6. The pooled average blends the sharply different responses that prevail across the full range of QoG in the sample. Only by evaluating the interaction away from that average, at the 25th and 75th percentiles, does the faint signal in Figure 4 resolve into the contrast shown here. That contrast is constrained to be linear in QoG: the specification forces the marginal response to move at a constant rate with the index, so the two evaluated points trace a line rather than describe two separate groups.

Figure 6: Consolidation effect by QoG



Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. The continuous interaction is evaluated at the 25th (Low QoG) and 75th (High QoG) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Table 3: Marginal cumulative response by QoG group and horizon (baseline continuous specification)

Outcome	QoG group	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$
Absolute redistribution	High QoG	0.064^{**}	0.153^{***}	0.209^{***}	0.228^{**}	0.212	0.134
		(0.025)	(0.036)	(0.071)	(0.104)	(0.150)	(0.231)
	Low QoG	-0.012	0.009	0.036	0.031	0.068	0.050
		(0.011)	(0.022)	(0.035)	(0.050)	(0.075)	(0.079)
Relative redistribution	High QoG	0.093	0.211^{***}	0.246^{*}	0.240	0.204	0.183
		(0.060)	(0.081)	(0.134)	(0.169)	(0.229)	(0.338)
	Low QoG	-0.032	-0.007	0.030	0.038	0.082	0.064
		(0.023)	(0.043)	(0.052)	(0.075)	(0.121)	(0.125)

Note: marginal cumulative response to a one-percentage-point-of-GDP fiscal consolidation, $\beta^h + \gamma^h q$, evaluated at each horizon h and at the 25th (Low QoG) and 75th (High QoG) percentiles of the QoG distribution. Standard errors in parentheses, backed out from the 90% confidence interval under a normal approximation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Own elaboration.

The conceptual framework in Section 3 predicts this pattern. QoG is used throughout this thesis as a proxy for the state's fiscal capacity, and that capacity matters for redistribution through two distinct channels: the capacity to tax income progressively and enforce collection over a broad formal base, and the capacity to sustain a transfer and public-spending system large enough that a consolidation of a given nominal size actually moves the disposable income of a meaningful share of the population. Where either form of capacity is present, as in high-

QoG countries, a fiscal consolidation has real redistributive machinery to work through, whether on the tax side or the spending side. Where it is absent, a nominally identical consolidation is adjusting a tax system that cannot reach income progressively and a transfer and spending system too thin to move the disposable income of more than a small fraction of households, so redistribution does not respond, however large the underlying fiscal effort.

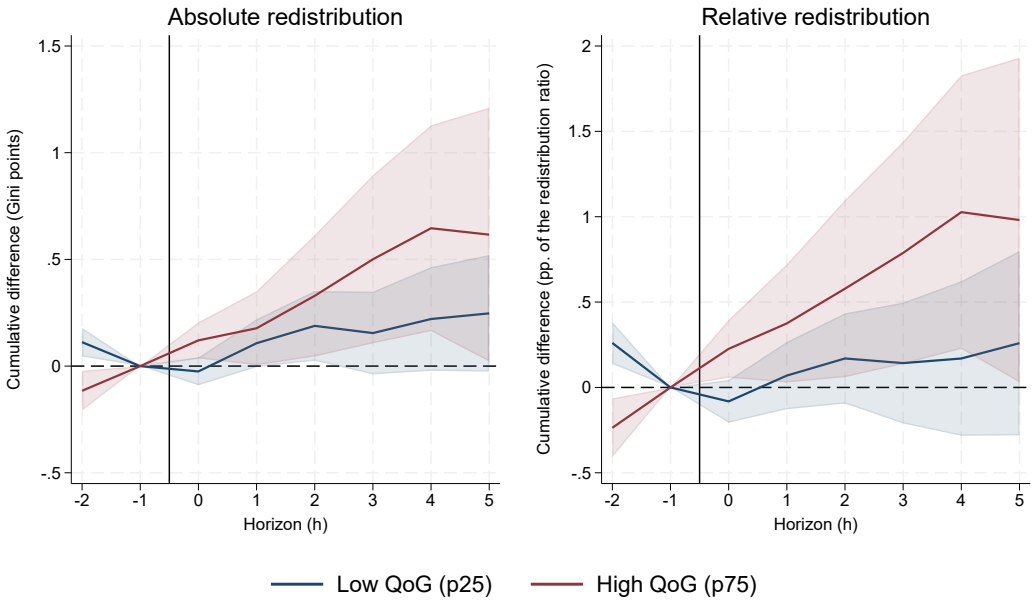
QoG correlates with income per capita, and the concern that it merely relabels the level of development deserves to be addressed directly. Two features of the results bear on it, though neither settles it. First, the QoG ranking does not reproduce a regional or development grouping. Italy, Portugal, and Spain fall in the low-QoG half of the sample while Chile falls in the high-QoG half (Table 8), so the index separates countries that a regional grouping keeps together. Second, substituting region for QoG does not reproduce the same estimates. As Appendix A.7 documents, the LAC group displays a significantly negative redistributive response where the low-QoG group displays none, so the two are empirically distinct rather than interchangeable labels. What this does not establish is that QoG is the operative dimension. A regional split that delivers the sharper contrast is, if anything, evidence that region carries heterogeneity beyond what QoG captures, and the two are correlated closely enough in this sample that the present design cannot separate their contributions. Adjudicating between them would require a specification that includes both, which is left for future work.

5.4 Conditional composition of adjustment

Having shown separately that composition matters on average (Figure 5) and that QoG conditions the overall redistributive response (Figure 6), the next question is whether these are really two separate findings or one and the same finding viewed from different angles. Is the composition effect documented above itself concentrated in high-QoG countries, in the same way the average effect turned out to be? The remainder of this subsection answers that question directly, re-estimating the QoG-conditional interaction separately for tax-based and expenditure-based consolidations.

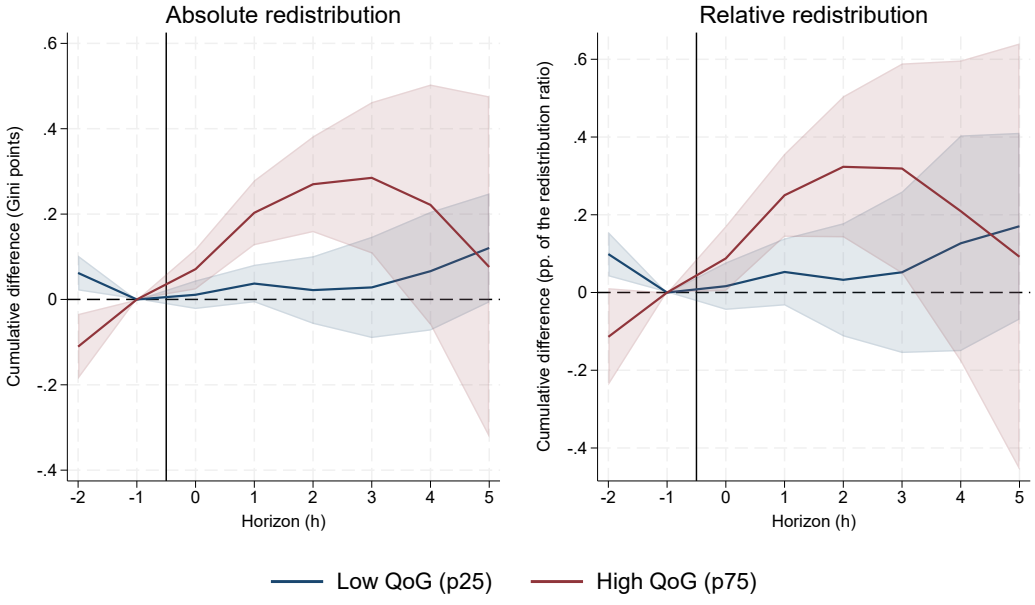
The answer is yes, with an important qualification, consistent with Hypothesis 2. Both types of consolidation reproduce the same QoG split documented above, with redistribution rising significantly for high-QoG countries over at least the early-to-medium run and showing no sustained response for low-QoG countries (Figures 7 and 8). The composition result from Figure 5 is therefore not a separate finding so much as a manifestation of the same state-capacity mechanism, present in both types of consolidation but concentrated more heavily in one of them. The qualification lies in that concentration, because the two cases differ sharply in strength. Tax-based consolidations in high-QoG countries produce a larger and more persistent redistribution response, remaining significant across almost the entire five-year post-shock horizon, while the expenditure-based response, though still significant over the medium run, plateaus by around $h = 3$ and loses precision at longer horizons as its confidence interval widens.

Figure 7: Tax-based consolidation by QoG



Note: Impulse response function of a fiscal consolidation interacted by QoG and Tax-based dummy on redistribution (absolute and relative) as cumulative difference. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. Tax-based consolidations defined as consolidations where the increase in government revenue is greater than the decrease in government spending caused by government austerity. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Figure 8: Expenditure-based consolidation by QoG



Note: Impulse response function of a fiscal consolidation interacted by QoG and Expenditure-based dummy on redistribution (absolute and relative) as cumulative difference. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. Expenditure-based consolidations defined as consolidations where the decrease in government spending is greater than the increase in government revenue caused by government austerity. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Read through the two channels laid out in Section 3, the progressivity channel appears to be the quantitatively stronger of the two mechanisms through which state capacity shapes the redistributive response to a consolidation, relative to the state-size channel operating through the scale of transfers and public spending. A tax increase in a high-capacity state reaches income directly and progressively, an effect that can compound for as long as the tax measure remains in place, consistent with the persistently rising response observed across the full window in Figure 7. An expenditure cut, even where the underlying transfer and spending system is large, still depends on that cut translating into lower disposable income for specific households. That link is arguably more exposed to fiscal or political reversal as the consolidation matures, consistent with the fading precision observed at the longer horizons in Figure 8. That expenditure-based consolidations widen redistribution at all, despite the regressive direct incidence of transfer and spending cuts emphasized in Section 3, is consistent with the market-income channel described there. Where the transfer and spending system is large, it continues to cushion disposable income even as it is being trimmed, so that market income can deteriorate by more than disposable income does and the wedge between the two widens. The Gini series are too imprecise here to confirm that decomposition directly, since neither moves significantly on its own (Figure 3), so this reading is an interpretation consistent with the framework rather than a separately established result. Table 4 summarizes the sign and significance of every specification presented in this section, at a representative medium-run horizon.

Table 4: Summary of redistribution effects across specifications ($h = 3$)

Specification	Group	Abs. redistribution	Rel. redistribution
Average effect (unconditional)	n/a	+	N
Composition (unconditional)	Tax-based	+	+
	Expenditure-based	+	N
QoG-conditional effect	Low QoG	N	N
	High QoG	+	N
Tax-based × QoG	Low QoG	N	N
	High QoG	+	+
Expenditure-based × QoG	Low QoG	N	N
	High QoG	+	+

Note: qualitative summary of the sign and statistical significance of each specification's effect on redistribution at a representative medium-run horizon ($h = 3$), read from the 90% confidence intervals in Figures 4, 5, 6, 7, and 8. "+" denotes a significant positive effect; "N" denotes no effect significant at the 90% level. Source: Own elaboration.

These four steps, taken together, trace a single argument rather than four separate findings. The pooled average effect is uninformative because it blends two populations whose responses differ not in degree but in kind. High-QoG countries display a clear, if temporally bounded, redistributive response to consolidation, while low-QoG countries display none, under every decomposition considered here. Composition compounds this picture rather than complicating it. Tax-based consolidations are the stronger redistributive instrument, but that strength is itself concentrated in high-QoG countries, consistent with a progressivity channel that requires the administrative capacity to collect on a progressive schedule in the first place. Across all four steps, the state's fiscal capacity does more to determine whether austerity redistributes than either the size or the nominal composition of the adjustment.

6 Robustness

Appendix A subjects the central result, that redistribution responds to fiscal consolidation only in high-QoG economies, to ten checks, organized here around three concerns rather than ten separate exercises.

The first concern is measurement. The continuous QoG interaction used throughout the main results is one modeling choice among several, so Appendix A.1 repeats the analysis using median and decile-based cutoffs instead, Appendix A.2 replaces the equal-weight WGI average with a principal-component index, and Appendix A.9 tests each of the four WGI components on its own rather than pooled into a single governance measure.

The second concern is identification. QoG could in principle respond to the consolidation itself rather than only condition it, so Appendix A.3 treats QoG as predetermined by using its value lagged one period. Appendix A.4 checks for a pre-existing divergence between high- and low-QoG countries at the earliest horizon the local projection window allows, and Appendix A.6 controls for a country- or region-specific linear trend to separate such a divergence from anticipation proper. Appendix A.5 drops the 2008–2009 window entirely, since the financial crisis could otherwise dominate a pooled panel spanning 1996–2019.

The third concern is whether the QoG split, and the sample underlying it, is doing the work rather than some other feature of the data. Appendix A.7 substitutes the OECD–LAC partition used in the descriptive motivation for the QoG split, Appendix A.8 re-estimates the headline specification dropping one country at a time, and Appendix A.10 adds an output-gap control to rule out that the QoG-conditional result is instead picking up the business cycle.

Across this battery, the divergence between high- and low-QoG economies documented in Section 5 survives every change of specification, though not the placebo. It holds under the decile-based cutoffs, every alternative weighting and governance component, under a predetermined rather than contemporaneous QoG, outside the financial crisis window, under leave-one-country-out resampling, and controlling for the output gap; the region check reproduces the high-capacity side of the result while revealing a distinct negative response in LAC (Appendix A.7). Two qualifications remain. The median-split cutoff produces a delayed positive response among low-QoG countries at longer horizons, which may reflect the pooling of near-median countries or genuine nonlinearity in QoG that the linear interaction cannot represent, and relative redistribution is significant at the medium-run horizon in only two of the four specifications reported in Table 6, while absolute redistribution is significant in all four. A third qualification is different in kind. The placebo test in Appendix A.4 shows that the two groups were already diverging in redistribution two years before the recorded consolidation, and in the same direction as the estimated effect. That divergence survives country- and region-specific linear trend controls in seven of eight cells (Appendix A.6), so it is not a simple secular trend, but no check reported here rules out that part of the post-shock response continues a movement already under way. The early-to-medium-run divergence is therefore the robust core of the result; claims about its persistence at the longest horizons, about relative redistribution specifically, and about reading the divergence as caused by the consolidation, call for more caution.

7 Limitations

This thesis is not without its limitations. First, the heterogeneity dimension chosen, QoG, is itself a composite, perception-based index. The four WGI components it averages are built from expert assessments and surveys rather than from directly observable administrative data, and cross-country and cross-time comparability of such perceptions is not guaranteed. The equal-

weight average used as the baseline is itself a modeling choice, though Appendix A.2 shows the results are virtually unchanged under a data-driven (PCA) weighting. A second concern is reverse causality. A fiscal consolidation could plausibly affect contemporaneously measured QoG, for instance if a credible adjustment improves perceived government effectiveness, or if austerity itself erodes it politically. The predetermined-QoG check in Appendix A.3 addresses this specifically by using QoG lagged one period, and the results are qualitatively unchanged, but this does not rule out more persistent forms of reverse causality operating over longer horizons.

A further measurement limitation concerns what the redistribution measures can capture. The wedge between the market and the disposable Gini reflects direct taxes and cash transfers only; transfers in kind, chiefly public health and education, enter neither Gini. The fiscal-incidence literature shows that in LAC these in-kind components are the main redistributive instrument (Lustig, 2017), so the redistribution measured here understates total fiscal redistribution, and the understatement is largest in the economies where the cash-based system is thinnest. Cuts to health and education consequently reach the income distribution through channels this design registers only partially: contemporaneously through public-sector wages and employment, which enter market income and are captured by the market Gini, and over the longer run through human capital accumulation, which the five-year window cannot capture, as discussed at the end of this section. Indirect taxes are excluded on the same principle, so a consolidation achieved through VAT, the margin low-capacity states rely on most, leaves no trace in the measured wedge, and part of the absent low-QoG response may be a measurement artifact rather than an economic null. SWIID also imputes the market Gini where surveys record net rather than gross incomes, as most LAC surveys do, so in that part of the sample the redistribution measure is partly model-generated rather than observed.

The classification of episodes by composition raises a separate issue. An episode is labelled expenditure-based when $\Delta G_{i,t} > \Delta T_{i,t}$ and tax-based when the reverse holds, with both components signed so that a positive value contributes to the adjustment. Twelve of the 191 episodes carry a negative net size, because one component reversed while the other was unchanged or moved by less, and in those cases the label records which component reversed rather than which instrument did the work: a net tax cut alongside unchanged spending is classified as expenditure-based, and a net spending increase alongside unchanged revenue as tax-based. None of the twelve exceeds one percent of GDP in absolute size, but the compositional results in Section 5 are best read as describing the dominant margin of adjustment rather than a clean partition of the sample.

In terms of expenditure and government revenue, each country has its own particularities in how these tools interact with the broader economy. Factors such as the size of the informal sector, the degree of financial development, the openness of the economy, and the credibility of fiscal institutions can all condition the effectiveness of fiscal policy, meaning that aggregate or pooled results may mask significant cross-country differences in the transmission mechanisms at play.

A further concern is anticipation. If agents foresee a consolidation, they may adjust their behavior before the episode is formally recorded, biasing the estimation. This is especially relevant under the narrative approach, in the hypothetical case that policy announcements preceded implementation by one or more years. A placebo test on the earliest pre-shock horizon that can be estimated within the local projection window (Appendix A.4) finds a statistically significant pre-trend in the redistribution measures. This could reflect an anticipation effect proper, or a broader secular trend, such as the inequality decline experienced by several LAC countries over the 2000s and 2010s, that predates and is unrelated to the specific consolidation

episode; the current evidence cannot fully distinguish between the two. The trend-controlled check in Appendix A.6 finds that the pre-shock divergence survives controlling for a country- or region-specific linear secular trend in seven of eight cells, evidence against a purely linear secular-trend explanation, though this does not on its own establish anticipation as the cause. The redistribution results should be read with this caveat in mind.

Finally, the channels through which a consolidation affects redistribution may themselves operate on different timelines. Tax changes and cuts to current transfers or public wages tend to have immediate effects on disposable income, while cuts to public services such as education or healthcare affect household welfare more gradually, through their effect on productivity and human capital accumulation over the medium to long run. This temporal heterogeneity means that the five-year window used throughout may not be sufficient to fully capture slower-moving channels, so the redistributive response documented here should be read as what materializes within that horizon, not necessarily its full long-run magnitude.

8 Conclusions

This thesis set out to ask whether fiscal consolidations redistribute income, and where. Pooling 30 OECD and LAC economies, the answer is conditional rather than unconditional. Consolidation leaves the pooled average Ginis statistically unchanged and moves redistribution only weakly and briefly, and only once QoG is allowed to condition the response does a clear pattern emerge. Redistribution rises after a consolidation in high-QoG economies, by roughly 0.23 Gini points per percentage point of GDP of consolidation by the third year, driven disproportionately by tax-based adjustment, while low-QoG economies show no comparable response in the baseline specification. This should be read alongside the placebo evidence in Appendix A.4: the two groups were already diverging before the shock, in the same direction as the estimated effect.

None of this should be read as a distributional endorsement of austerity. The evidence here does not overturn the literature's central finding that consolidations are, on average, disequalizing (Ball et al., 2013; Woo et al., 2013; Heimberger, 2020), but neither does it corroborate it. The pooled estimates leave both Gini levels statistically indistinguishable from zero, and so support no claim about the direction in which the level of inequality moved in this sample. What the data do identify is a widening of the gap between the two Ginis in high-QoG economies, which is consistent with a fiscal system absorbing part of a shock, not with a consolidation that leaves the income distribution more equal than it found it.

The result carries a specific policy implication. The distributional consequences of austerity are not primarily a function of how large the adjustment is, or even of whether it leans on taxes or spending, but of the administrative capacity of the state carrying it out. A tax-based consolidation only widens redistribution where the tax administration can already levy income taxes on a progressive schedule; the same nominal design choice in a low-capacity state does not appear to move redistribution at all. This qualifies the distributional case for preferring tax-based over expenditure-based consolidation. That case is conditional on state capacity that cannot simply be assumed, and where it is absent, that capacity has to be built first: a broader tax base, an administration able to assess and enforce what the base implies, and a transfer system whose coverage reaches the households at the bottom of the distribution. Until then, the instrument choice has little purchase on the margin measured here.

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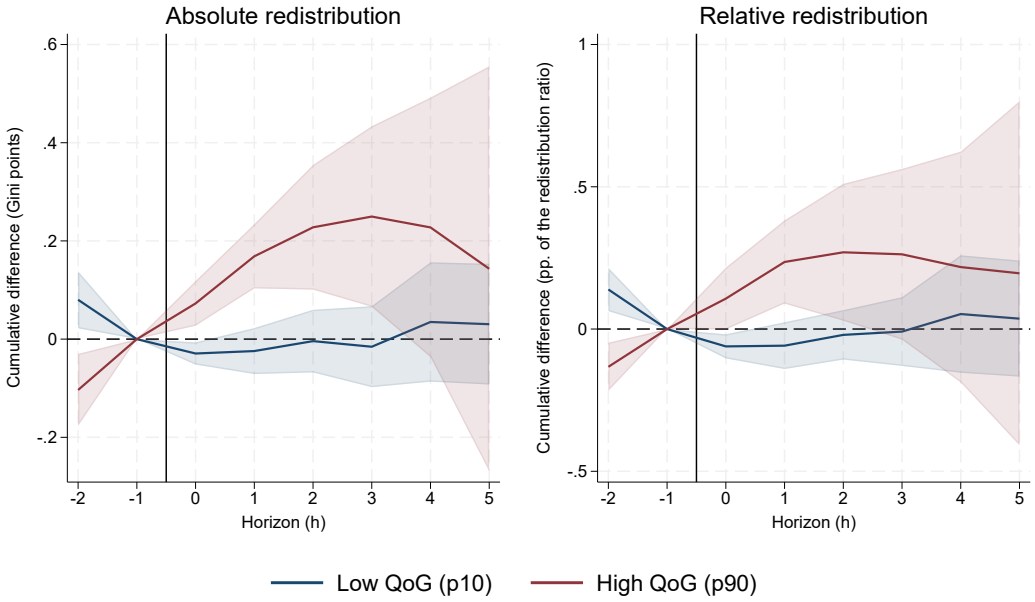
A Robustness Estimations

A.1 Alternative governance cutoffs

The baseline continuous interaction is evaluated at the 25th and 75th percentiles of QoG. As a check that this particular choice of cutoffs is not driving the result, this section considers two alternatives: evaluating the same continuous interaction at the first and last deciles (10th and 90th percentiles), a more extreme contrast between low- and high-QoG countries; and replacing the continuous interaction altogether with a binary split at the median, so that every country-year is classified as either Low or High QoG.

Widening the contrast to the first and last deciles (Figure 9) broadly reproduces the baseline pattern. Absolute redistribution rises significantly for high-QoG (p90) countries from the year of the shock, and relative redistribution from $h = 1$, consistent with Figure 6. Low-QoG (p10) countries are not simply flat here, however. Both redistribution measures move significantly negative on impact ($h = 0$), before settling into the same statistically insignificant response found at low QoG elsewhere in this appendix.

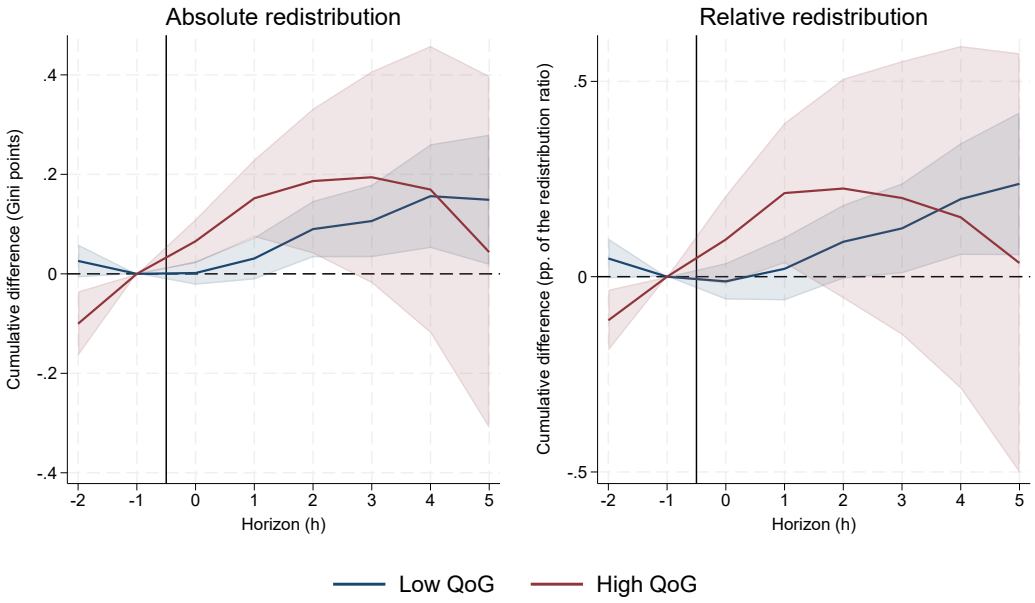
Figure 9: Consolidation effect by QoG using first and last decile



Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. The continuous interaction is evaluated at the 10th (Low QoG) and 90th (High QoG) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

The median split (Figure 10) departs from the baseline pattern more substantially. High QoG rises significantly over $h = 0$ to $h = 2$, as elsewhere, but low QoG is not null. It turns significantly *positive* from around $h = 2$ (absolute redistribution) or $h = 4$ (relative redistribution) onward, and by $h = 5$ exceeds the by-then-insignificant high-QoG estimate for both measures. The "crossing" at $h = 5$ therefore reflects low QoG becoming significant on a delay, not high QoG fading into noise while low QoG stays flat, as in the continuous specification.

Figure 10: Consolidation effect by QoG regime



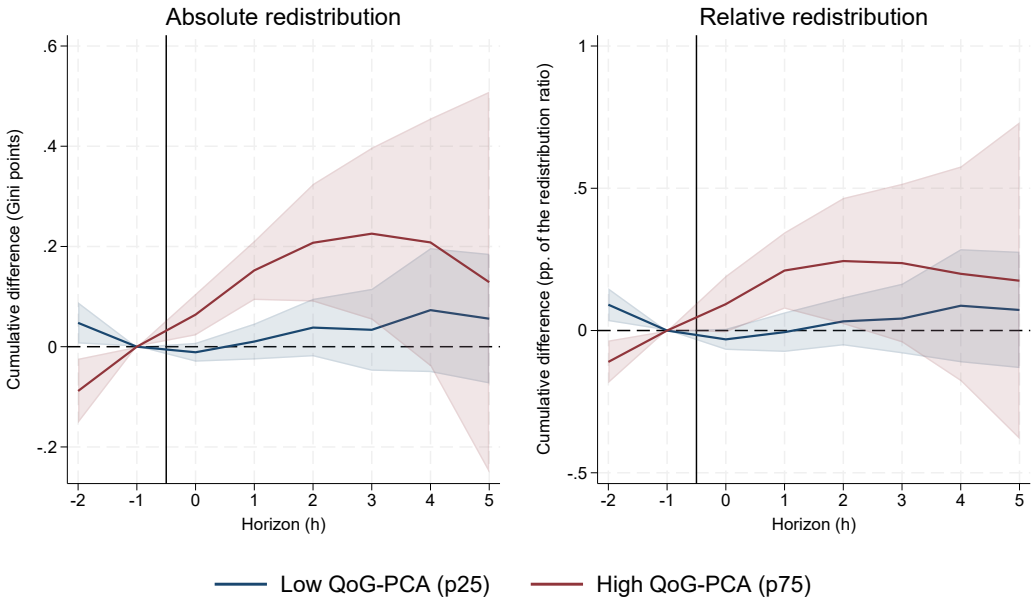
Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. Low QoG represents below median, while High QoG is above median. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Two readings are available. One is a property of the median split itself: pooling every below-median country-year into a single "low" group, rather than evaluating the continuous interaction at a fixed low percentile, mixes in countries close to the median where the state-capacity mechanism should already be partly active, pulling the group average away from zero at longer horizons. The other is that the response is genuinely nonlinear in QoG, in which case the binary split is picking up curvature that the linear interaction is constrained not to represent. The evidence here cannot separate the two, and on the second reading the finding is a limitation of the baseline specification rather than a defect of this check. Read together with the sample-sensitivity and country-trend checks elsewhere in this appendix, the QoG-conditional divergence is most reliably estimated in the early-to-medium run, and the long-horizon behavior of the low-QoG group is the part of the picture that depends most on how QoG is allowed to enter the specification.

A.2 Alternative governance index

The baseline QoG index averages the four WGI estimates with equal weights (0.25 each). As an alternative, a data-driven index is built from the first principal component of the same four series, letting their relative weights reflect the correlation structure of the data instead of an ad-hoc choice.

Figure 11: Consolidation effect by QoG, PCA-based index



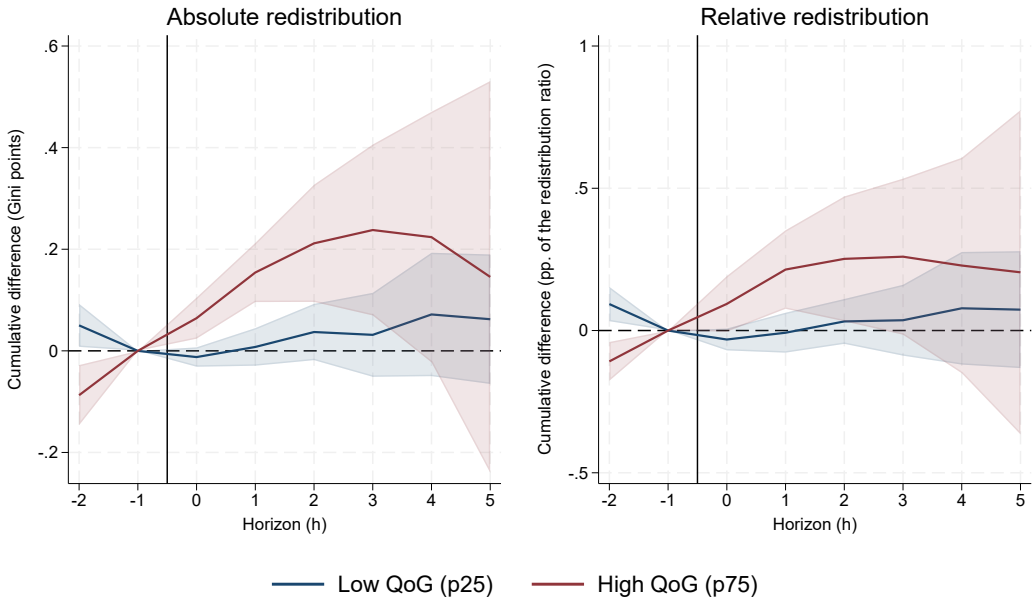
Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference. QoG defined as the first principal component of the estimates of: control of corruption (loading 0.501), government effectiveness (0.500), rule of law (0.504), and regulatory quality (0.495), instead of their simple average. The near-identical loadings reflect how highly correlated the four WGI dimensions are with one another. The continuous interaction is evaluated at the 25th (Low QoG) and 75th (High QoG) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

The results under the PCA-based index (Figure 11) are virtually unchanged relative to the equal-weight baseline (Figure 6), which is unsurprising given how highly correlated the four WGI dimensions are with one another; the choice of aggregation rule is therefore not driving the main finding.

A.3 Endogeneity

This check addresses the possibility that a fiscal consolidation itself affects contemporaneously measured QoG, for instance if a credible adjustment improves perceived government effectiveness, or if austerity erodes it politically, by replacing the contemporaneous interaction with QoG lagged one period, predetermined relative to the shock.

Figure 12: Consolidation effect by predetermined QoG



Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. QoG enters the interaction predetermined at $t-1$. The continuous interaction is evaluated at the 25th (Low QoG) and 75th (High QoG) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

The predetermined specification (Figure 12) reproduces the baseline pattern almost exactly. Both redistribution measures rise significantly for high-QoG countries over the same horizons as in Figure 6, and remain insignificant for low-QoG countries. Using QoG measured before the shock therefore does not change the qualitative or quantitative picture, suggesting that reverse causality from the consolidation to contemporaneously measured QoG is not what is driving the baseline result.

A.4 Pre-trends

A further concern for any local projection is that the estimated response may reflect a divergence between groups that was already under way before the shock, rather than an effect caused by it. Under the cumulative transformation used throughout ($\Delta Y_{i,t+h} = Y_{i,t+h} - Y_{i,t-1}$), the response at $h=-1$ is zero by construction (the normalization period) and therefore uninformative for this purpose; the only pre-shock horizon that is actually estimated within the $h \in \{-2,\dots,5\}$ window is $h=-2$. Table 5 reports the marginal cumulative response at $h=-2$, $\beta^h + \gamma^h q$, evaluated at the 25th and 75th percentiles of QoG, for each outcome.

Table 5: Pre-shock ($h = -2$) coefficient, by outcome and QoG group

Outcome	QoG group	Coefficient	SE	p-value
Absolute redistribution	Low	0.048^{*}	0.025	0.055
	High	-0.088^{**}	0.039	0.024
Relative redistribution	Low	0.092^{***}	0.035	0.009
	High	-0.110^{**}	0.046	0.016

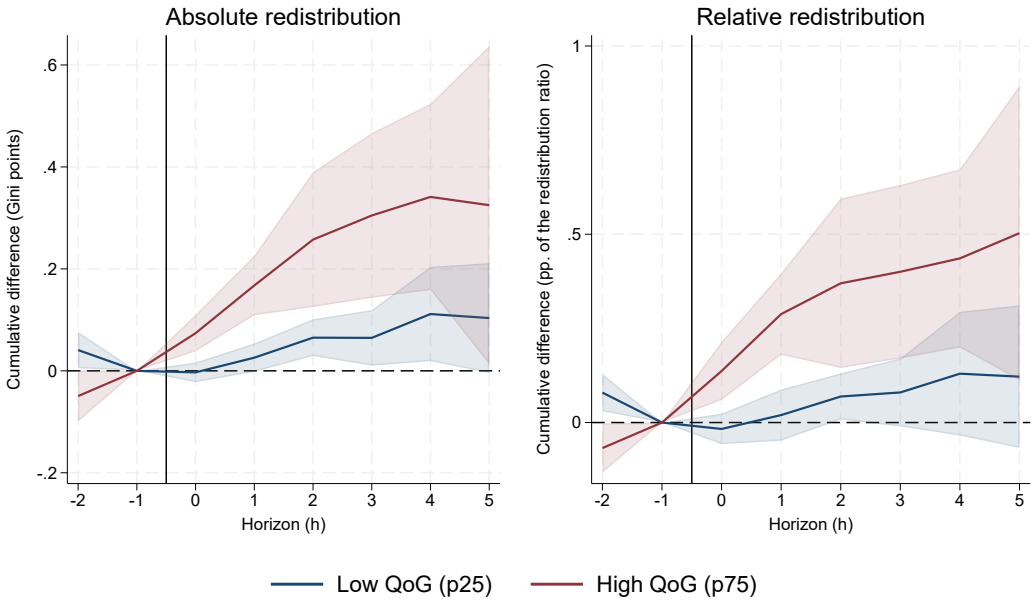
Note: Marginal cumulative response to a one-percentage-point-of-GDP fiscal consolidation, $\beta^h + \gamma^h q$, at horizon $h = -2$ (two years before the consolidation), from the same continuous specification as Figure 6, evaluated at the 25th (Low) and 75th (High) percentiles of QoG. Standard errors backed out from the 90% confidence interval reported by the local projection under a normal approximation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Own elaboration.

Both redistribution measures show a significant pre-shock coefficient for both QoG groups. The dependent variable at $h = -2$ is $Y_{i,t-2} - Y_{i,t-1}$, so the negative high-QoG coefficient in Table 5 means redistribution was already rising over the two years before the consolidation, and the positive low-QoG coefficient means it was falling. The two groups were therefore diverging in the same direction as the post-shock estimates before the shock occurred, which is the pattern a pre-existing trend would produce. This is the most serious threat to identification in the analysis, and the redistribution results should be read with it in mind.

A.5 Global financial crisis exclusion

Two of the sample years, 2008 and 2009, correspond to the global financial crisis (GFC), a period of unusually large and globally synchronized swings in output, government revenue, and possibly measured QoG. This check re-estimates the baseline continuous interaction dropping any observation whose own year falls in 2008 or 2009, to verify the headline result is not an artifact of the crisis contaminating either the timing of consolidations or the measurement of QoG in those years.

Figure 13: Consolidation effect by QoG, excluding 2008–2009



Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference, excluding observations whose own year falls in 2008 or 2009. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. The continuous interaction is evaluated at the 25th (Low QoG) and 75th (High QoG) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

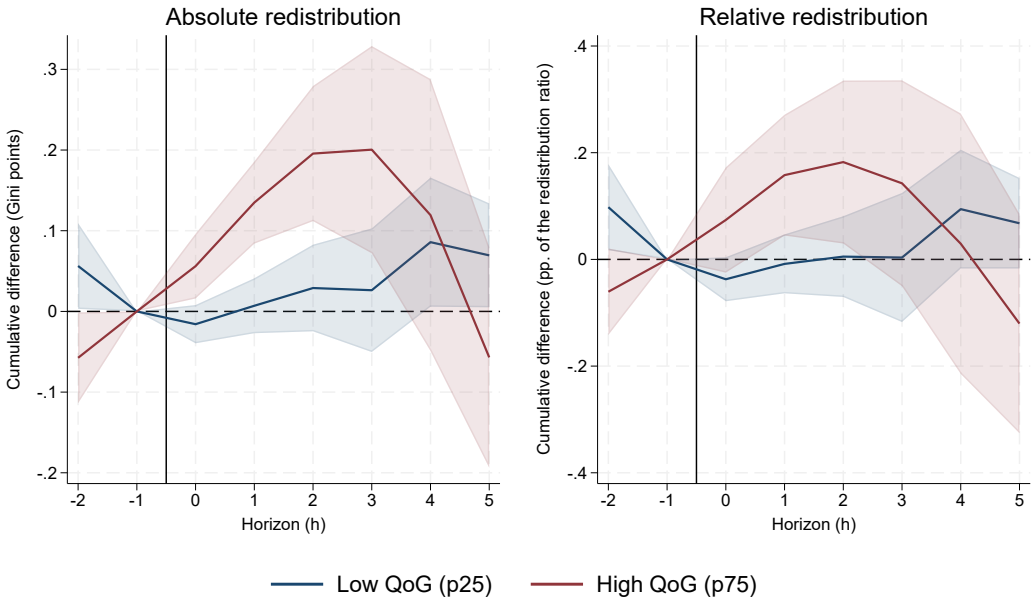
Excluding the crisis years (Figure 13) does not weaken the baseline pattern, and if anything strengthens it. Both redistribution measures rise significantly for high-QoG countries, more strongly and more persistently than in the full sample, remaining significant through $h = 5$, while low-QoG countries remain insignificant throughout. The headline result is therefore not an artifact of the 2008–2009 crisis.

A.6 Country- and region-specific trends

The pre-trends check in Appendix A.4 finds a statistically significant divergence between low- and high-QoG countries already present at $h = -2$, before the consolidation. One candidate explanation is that this reflects a country- or region-level secular trend in redistribution, rather than genuine anticipation of the consolidation, for instance the gradual inequality decline observed in several LAC countries over the 2000s and 2010s. This check re-estimates the baseline continuous interaction with a country-specific linear time trend added as an additional control, on top of the country fixed effects already included, and separately with a coarser region-specific trend (a common linear trend interacted with the LAC dummy).

The shape of the response changes once a country-specific trend is controlled for (Figure 14). The high-QoG effect still rises and becomes significant over the early-to-medium run, but it peaks around $h = 2$ to $h = 3$ and fades back toward zero by $h = 5$, rather than continuing to widen. Table 6 makes this precise, comparing the high-QoG coefficient at $h = 3$ and $h = 5$ across the baseline specification, the GFC exclusion (Appendix A.5), and both trend-controlled specifications.

Figure 14: Consolidation effect by QoG, controlling for country-specific linear trends



Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference, additionally controlling for a country-specific linear time trend. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. The continuous interaction is evaluated at the 25th (Low QoG) and 75th (High QoG) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Table 6: High-QoG coefficient at $h = 3$ and $h = 5$, across specifications

Outcome	Horizon	Baseline	No GFC	Country trend	Region trend
Absolute redistribution	$h = 3$	0.228^{**}	0.305^{***}	0.200^{**}	0.233^{**}
		(0.104)	(0.098)	(0.078)	(0.091)
	$h = 5$	0.134	0.325^{*}	-0.057	0.078
		(0.231)	(0.190)	(0.083)	(0.190)
Relative redistribution	$h = 3$	0.240	0.400^{***}	0.143	0.242^{*}
		(0.169)	(0.140)	(0.117)	(0.147)
	$h = 5$	0.183	0.503^{**}	-0.120	0.097
		(0.338)	(0.238)	(0.125)	(0.275)

Note: marginal cumulative response to a one-percentage-point-of-GDP fiscal consolidation, $\beta^h + \gamma^h q$, evaluated at the 75th percentile of QoG, at horizons $h = 3$ and $h = 5$, under four specifications of the baseline continuous interaction. Standard errors in parentheses, backed out from the 90% confidence interval under a normal approximation. The corresponding low-QoG coefficient is not significant at the 90% level in nearly every cell across all four specifications, with only two scattered exceptions. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Own elaboration.

The $h = 3$ result is robust: the high-QoG coefficient on absolute redistribution is significant across all four specifications, and relative redistribution is significant in two of the four. The $h = 5$ result is not. Once either trend control is added, the coefficient collapses toward zero and,

in the country-specific case, turns negative, though never significantly so. The genuinely trend-robust part of the headline result is therefore best described as an early-to-medium-run effect, concentrated around $h = 1$ to $h = 3$, rather than a monotonically widening one that persists to the end of the estimation window.

A follow-up question is whether the same secular trend also explains the significant pre-shock coefficient documented in Appendix A.4. Table 7 repeats that placebo test at $h = -2$ under both trend-controlled specifications. The pre-shock coefficient survives the trend controls in seven of the eight cells; only relative redistribution for the high-QoG group loses significance once a country-specific trend is added. This is the opposite of what a purely secular-trend explanation would predict. If the $h = -2$ divergence were simply the same trend that erodes the $h = 5$ result, controlling for that trend should have made it disappear. Since it largely does not, the pre-shock divergence documented in Appendix A.4 reflects something other than a simple linear secular trend, and remains a genuine limitation of the identification strategy rather than an artifact resolved by this check.

Table 7: Pre-shock ($h = -2$) coefficient under trend controls

Outcome	QoG group	Baseline	Country trend	Region trend
Absolute redistribution	Low	0.048^{*}	0.056^{*}	0.047^{*}
		(0.025)	(0.032)	(0.024)
	High	-0.088^{**}	-0.057^{*}	-0.093^{**}
		(0.039)	(0.034)	(0.038)
Relative redistribution	Low	0.092^{***}	0.098^{**}	0.085^{**}
		(0.035)	(0.049)	(0.034)
	High	-0.110^{**}	-0.061	-0.117^{**}
		(0.046)	(0.049)	(0.046)

Note: marginal cumulative response to a one-percentage-point-of-GDP fiscal consolidation, $\beta^h + \gamma^h q$, at horizon $h = -2$, evaluated at the 25th (Low) and 75th (High) percentiles of QoG, under the baseline specification and both trend-controlled specifications. Standard errors in parentheses, backed out from the 90% confidence interval under a normal approximation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Own elaboration.

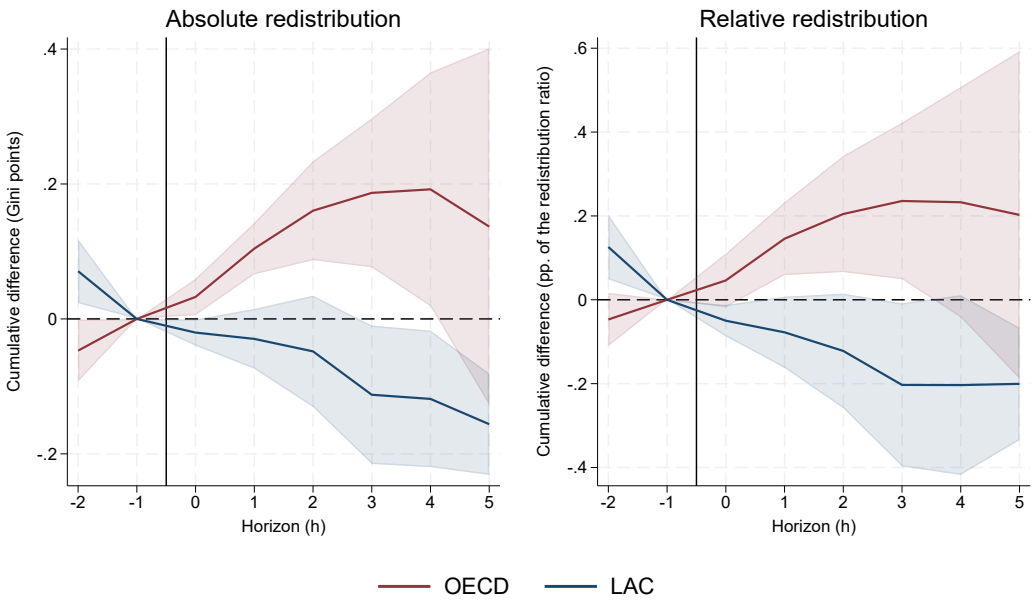
A.7 Region as a competing partition

As a competing partition to QoG, this check splits the sample by region instead, comparing OECD countries with Latin American and Caribbean (LAC) countries. Region correlates strongly with QoG in this sample (OECD countries are disproportionately high-QoG and LAC countries disproportionately low-QoG), but the two are not identical, so this also serves as an informal test of whether QoG or some broader regional factor is doing the work.

The regional split (Figure 15) reproduces the direction of the baseline result. OECD countries show a significant increase in both redistribution measures, comparable in timing and magnitude to the high-QoG group in Figure 6, but the LAC estimate is not merely insignificant, as the low-QoG group is in the continuous specification. It is significantly negative for both measures from around $h = 2$ onward. Fiscal consolidations in LAC countries are therefore associated with a measurable narrowing of redistribution, a sharper and qualitatively different result from the null effect found for low-QoG countries elsewhere in this appendix. This suggests that region captures something beyond QoG alone, such as weaker automatic stabilizers

or narrower social-protection coverage specific to the LAC experience, that is worth keeping in mind when interpreting the low-QoG side of the baseline result.

Figure 15: Consolidation effect by region



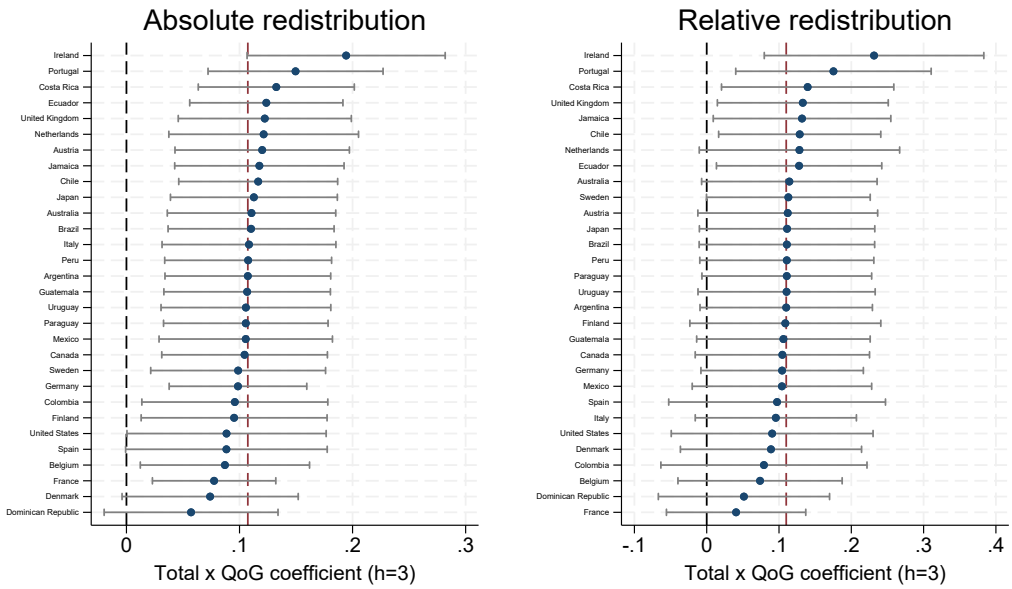
Note: Impulse response function of a fiscal consolidation interacted by a region dummy (LAC vs. OECD) on redistribution (absolute and relative) as cumulative difference. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

A.8 Sample sensitivity

To verify that no single country in the panel drives the baseline result, the consolidation $\times$ QoG interaction coefficient is re-estimated leaving out one country at a time, evaluated at a representative medium-run horizon ($h = 3$).

The coefficient (Figure 16) retains its sign and remains of comparable magnitude to the full-sample estimate (dashed line) across all leave-one-out samples and for every outcome, indicating that the baseline result is not driven by any single country in the panel.

Figure 16: Consolidation effect by QoG: leave-one-out by country



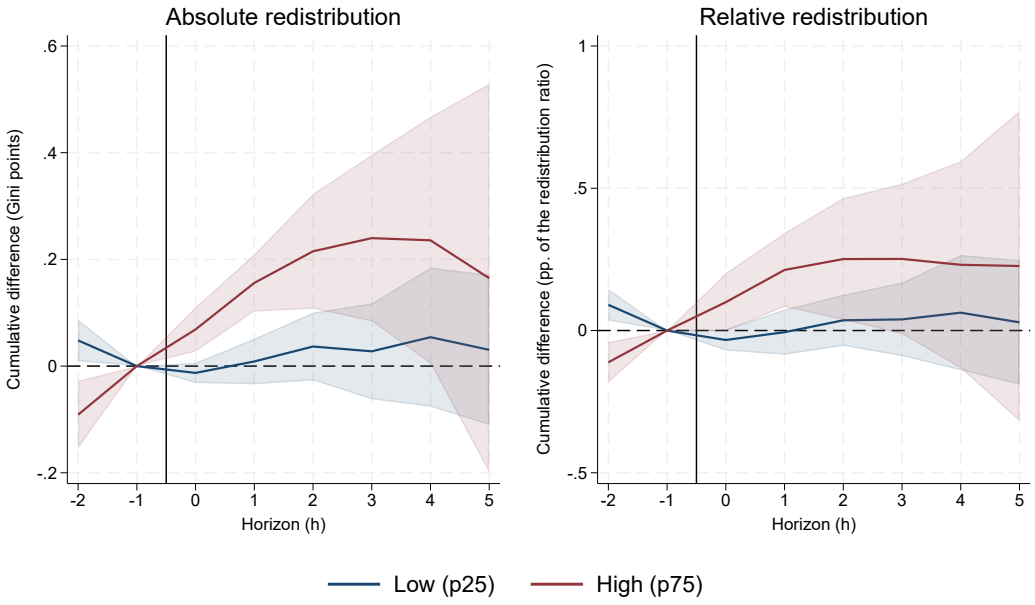
Note: Each dot reports the point estimate (with 90% confidence interval) of the coefficient on the interaction between the fiscal consolidation shock and QoG, for redistribution (absolute and relative), at horizon $h = 3$, re-estimated after excluding one country at a time from the estimation sample. Countries are ordered by the magnitude of the coefficient. The dashed vertical line marks the full-sample estimate; the solid vertical line marks zero. QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. Source: Own elaboration.

A.9 QoG components

The baseline QoG index averages four WGI dimensions with equal weights. This check re-estimates the continuous interaction separately for each of the four components (control of corruption, government effectiveness, rule of law, and regulatory quality) to verify that no single dimension is mechanically responsible for the baseline result.

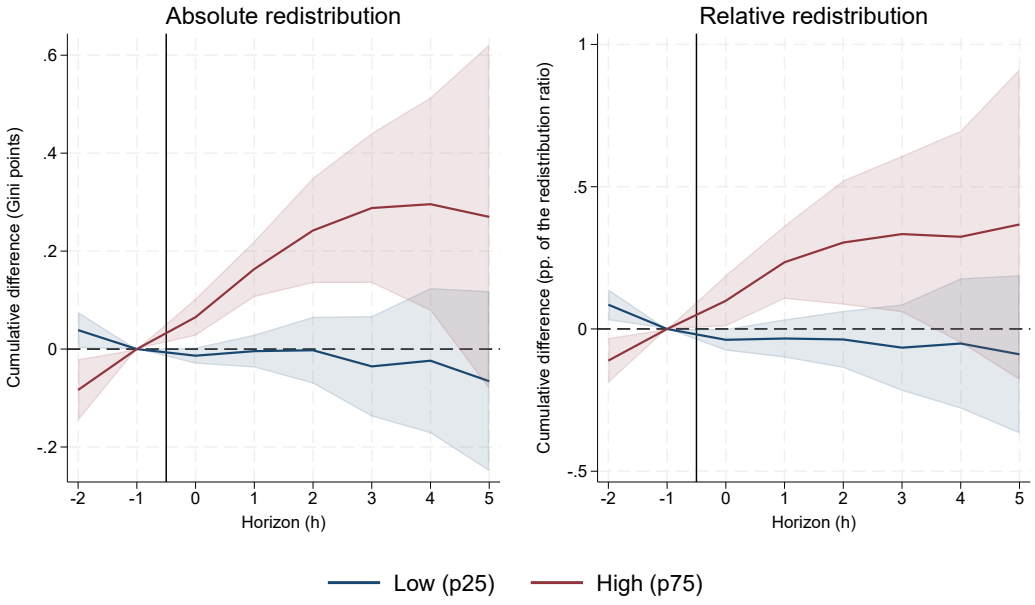
Control of corruption (Figure 17) and government effectiveness (Figure 18) reproduce the baseline pattern cleanly. Both redistribution measures rise significantly and persistently for countries scoring high on either dimension, with no significant response among low-scoring countries, and government effectiveness gives, if anything, the strongest and most persistent version of the result. Rule of law (Figure 19) and regulatory quality (Figure 20) reproduce the same divergence over the early-to-medium run, but it narrows steadily from around $h = 3$ onward and, for regulatory quality in particular, the two groups cross by $h = 5$, with the low-scoring group ending above the high-scoring one.

Figure 17: Consolidation effect by control of corruption



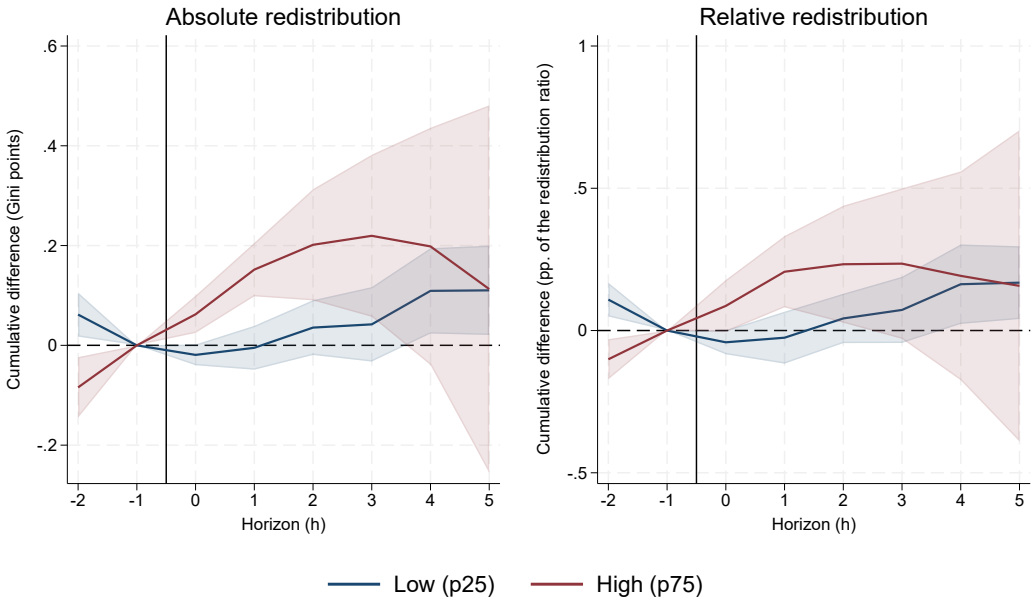
Note: Impulse response function of a fiscal consolidation interacted by the control of corruption estimate (WGI) on redistribution (absolute and relative) as cumulative difference. The continuous interaction is evaluated at the 25th (Low) and 75th (High) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Figure 18: Consolidation effect by government effectiveness



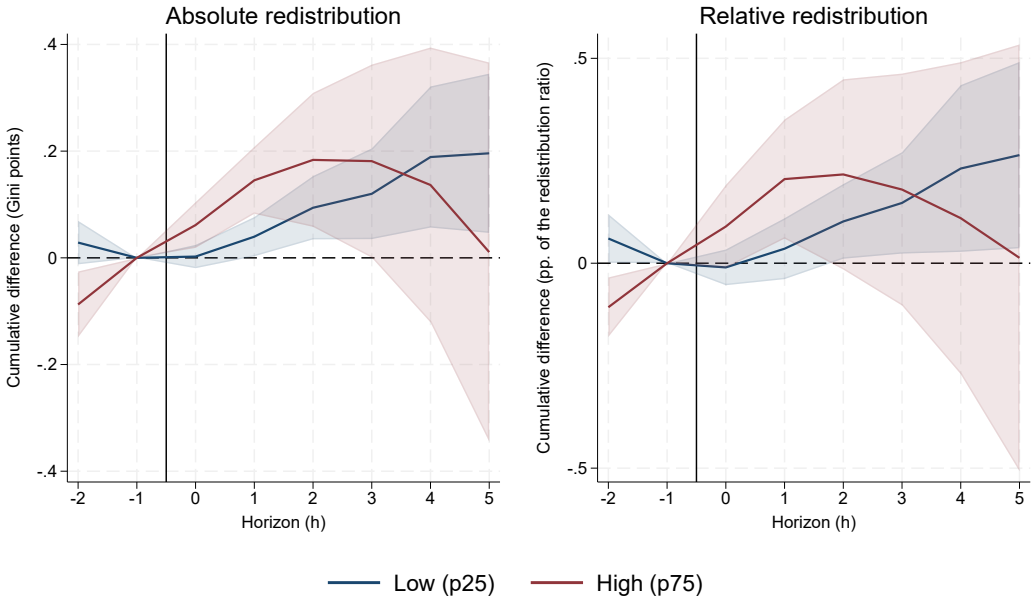
Note: Impulse response function of a fiscal consolidation interacted by the government effectiveness estimate (WGI) on redistribution (absolute and relative) as cumulative difference. The continuous interaction is evaluated at the 25th (Low) and 75th (High) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Figure 19: Consolidation effect by rule of law



Note: Impulse response function of a fiscal consolidation interacted by the rule of law estimate (WGI) on redistribution (absolute and relative) as cumulative difference. The continuous interaction is evaluated at the 25th (Low) and 75th (High) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Figure 20: Consolidation effect by regulatory quality



Note: Impulse response function of a fiscal consolidation interacted by the regulatory quality estimate (WGI) on redistribution (absolute and relative) as cumulative difference. The continuous interaction is evaluated at the 25th (Low) and 75th (High) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

The baseline result is therefore not an artifact of any single dimension, but it is more robust along the components capturing administrative and bureaucratic capacity (government effec-

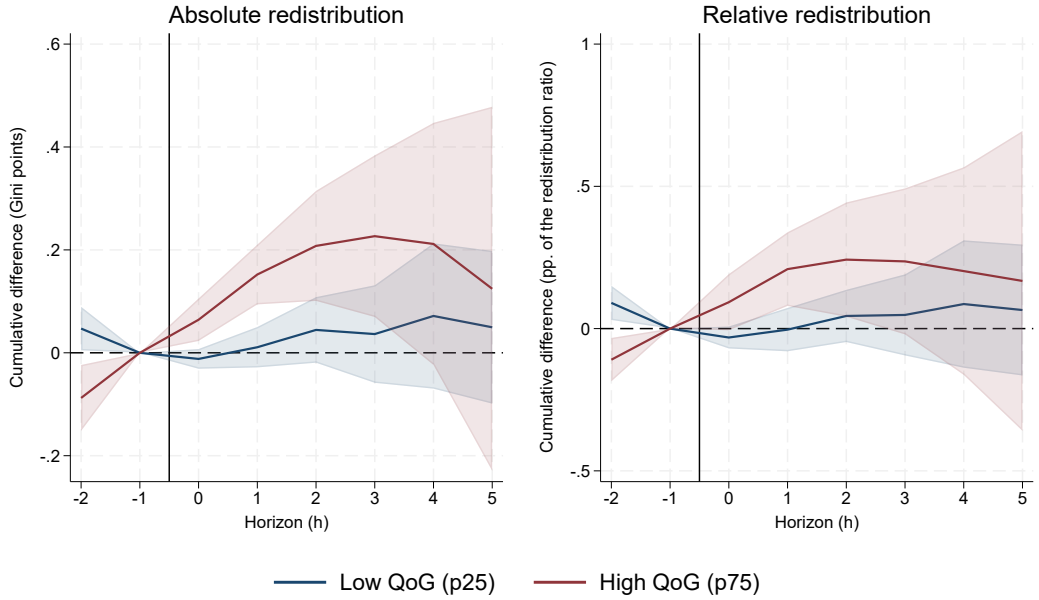
tiveness and control of corruption) than along those capturing the legal and regulatory environment, at least at the longest horizons considered.

A.10 Controlling for the output gap

As an additional check, the baseline continuous QoG interaction is re-estimated with the Hodrick–Prescott output gap (Hodrick and Prescott, 1997) added as a control, to verify the result is not confounded by where a country sits in its business cycle when a (narrative) consolidation occurs.

Controlling for the output gap (Figure 21) leaves the baseline pattern essentially unchanged. Both redistribution measures still rise significantly for high-QoG countries, closely tracking Figure 6 in both timing and magnitude, and remain insignificant for low-QoG countries. The result is therefore unlikely to reflect low- and high-QoG countries simply being at different points in the business cycle when a consolidation occurs.

Figure 21: Consolidation effect by QoG, controlling for the output gap



Note: Impulse response function of a fiscal consolidation interacted by QoG on redistribution (absolute and relative) as cumulative difference, additionally controlling for the output gap (Hodrick–Prescott filter on log real GDP). QoG defined as the simple average of the estimates of: control of corruption, government effectiveness, rule of law and regulatory quality. The continuous interaction is evaluated at the 25th (Low QoG) and 75th (High QoG) percentiles. Shaded areas represent 90% confidence intervals. Source: Own elaboration.

Across this battery of checks, the QoG-conditional divergence documented in the main results holds up well in the specifications that matter most, though the pre-shock divergence documented in Appendix A.4 remains a separate and more serious caveat. It survives excluding the financial crisis, weighting the governance index differently, treating QoG as predetermined, dropping any single country, isolating individual WGI components, and controlling for the output gap; the region check reproduces the high-capacity side of the result while revealing a distinct negative response in LAC. Two genuine exceptions temper this picture. The median-split cutoff, unlike the continuous interaction or the decile-based cutoffs, produces a delayed positive response in the low-QoG group at longer horizons, which may reflect the pooling of

near-median countries or genuine nonlinearity in QoG that the linear specification cannot represent. And relative redistribution is significant at the medium-run horizon in only two of the four specifications reported in Table 6, while absolute redistribution is significant in all four. The early-to-medium-run divergence between high- and low-QoG countries is therefore the most robust part of the headline result; claims about its persistence to the longest horizons, and about relative redistribution in particular, warrant more caution.

B Supplementary material

B.1 Local Projections and Vector Autoregressive Models

The use of Local Projections (LP) rather than a variation of a Vector Autoregressive Model (VAR) is justified as follows: the estimation of a VAR relies on the observed sample and delivers a global linear approximation that can be optimally constructed for short-run forecasting, even under model misspecification. However, impulse responses are built from forecasts at progressively longer horizons, so any specification error tends to accumulate over time. By contrast, the LP approach estimates a sequence of regressions in which the dependent variable is shifted forward to each specific horizon, making the estimation local to that horizon (hence the name) and yielding consistent impulse response coefficients, while also being comparatively robust to misspecification (Jordà, 2005).

Plagborg-Møller and Wolf (2021) show that LPs and VARs estimate the same impulse responses when the lag structures of both specifications are unrestricted, requiring only the nonparametric assumption that the data are weakly stationary, and that with a limited number of lags p included in the LP and VAR, the two impulse response functions approximate each other up to that horizon ($h \leq p$) without imposing parametric assumptions on the data generating process, but may differ for intermediate and long horizons ($h > p$). Following this, for the case of finite samples, there is a stark bias–variance trade-off between LP and VAR impulse response estimators. The least-squares LP estimator tends to have lower bias than the least-squares VAR estimator while also having a substantially higher variance (Li et al., 2024).

In this regard, LPs are based on single-equation estimation, which is particularly helpful when estimating the entire system is impractical because of limited data or model complexity. Because LPs operate equation by equation, they are well suited to settings with nonlinearities or state dependence, although careful interpretation is required. Moreover, LPs simplify both estimation and inference for key quantities of interest, including cumulative effects and multipliers. Finally, they offer a flexible framework that accommodates panel data as well as difference-in-differences and staggered event-study designs with heterogeneous treatment effects, albeit at the cost of the aforementioned bias–variance trade-off (Jordà and Taylor, 2025).

B.2 Fiscal consolidation episodes

Table 8 lists, for each country in the analysis sample (1996–2019, excluding Bolivia), the years in which it undertook a fiscal consolidation, as identified by Adler et al. (2024). Countries are ranked by QoG (the country average, reported alongside) and grouped into the Low- and High-QoG halves used in the descriptive figures, rather than by region; the estimations use the continuous index, evaluated at its 25th and 75th percentiles.

Table 8: Fiscal consolidation episodes by country identified by Adler et al. (2024), 1996–2019

	Country	ISO	Region	QoG	Years of Consolidation
Low QoG	Paraguay	PRY	LAC	−0.83	2001, 2003, 2004, 2005, 2006, 2014, 2016
	Ecuador	ECU	LAC	−0.74	2000, 2018, 2019
	Guatemala	GTM	LAC	−0.72	1996, 2000, 2002, 2012, 2013
	Dominican Republic	DOM	LAC	−0.52	2004, 2006, 2007, 2011, 2013
	Argentina	ARG	LAC	−0.30	1996, 1997, 2018, 2019
	Colombia	COL	LAC	−0.29	2000, 2003, 2011, 2012, 2015, 2016, 2017
	Peru	PER	LAC	−0.27	2002, 2003, 2011, 2012
	Mexico	MEX	LAC	−0.20	2010, 2014
	Brazil	BRA	LAC	−0.11	2015, 2017
	Jamaica	JAM	LAC	−0.03	1999, 2000, 2003, 2004, 2012, 2013, 2014
	Costa Rica	CRI	LAC	0.51	1996, 1997, 2016
	Italy	ITA	OECD	0.55	1996, 1997, 1998, 2004, 2005, 2006, 2007, 2010, 2011, 2012, 2013, 2014
	Uruguay	URY	LAC	0.74	1996, 2000, 2002, 2003, 2004, 2005, 2015
	Portugal	PRT	OECD	1.01	2000, 2002, 2003, 2005, 2006, 2007, 2010, 2011, 2012, 2013, 2014
	Spain	ESP	OECD	1.14	1996, 1997, 2010, 2011, 2012, 2013, 2014, 2017
High QoG	Japan	JPN	OECD	1.24	1997, 1998, 2003, 2004, 2005, 2006, 2007, 2014, 2015, 2019
	Chile	CHL	LAC	1.27	2003, 2004, 2008, 2014, 2015, 2016
	France	FRA	OECD	1.33	1996, 1997, 1999, 2000, 2011, 2012, 2013, 2014, 2015, 2018
	Belgium	BEL	OECD	1.41	1996, 1997, 2010, 2011, 2012, 2013, 2014
	United States	USA	OECD	1.51	1996, 1997, 1998, 2011, 2012, 2013
	Ireland	IRL	OECD	1.54	2009, 2010, 2011, 2012, 2013, 2014
	Germany	DEU	OECD	1.68	1997, 1998, 1999, 2000, 2003, 2004, 2006, 2007, 2011, 2012, 2013
	United Kingdom	GBR	OECD	1.69	1996, 1997, 1998, 1999, 2010, 2011, 2012, 2013, 2014, 2015
	Austria	AUT	OECD	1.71	1996, 1997, 2001, 2002, 2011, 2012, 2013, 2014
	Australia	AUS	OECD	1.76	1996, 1997, 1998, 1999
	Canada	CAN	OECD	1.79	1996, 1997, 2010, 2011, 2012, 2013, 2014
	Netherlands	NLD	OECD	1.90	2004, 2005, 2011, 2012, 2013
	Sweden	SWE	OECD	1.91	1996, 1997, 1998
	Denmark	DNK	OECD	2.01	2011, 2012, 2013
	Finland	FIN	OECD	2.04	1996, 1997, 2011, 2012, 2013, 2014, 2015, 2016

Source: Own elaboration based on data from the Worldwide Governance Indicators (WGI) and fiscal consolidation data from Adler et al. (2024).

B.3 Additional descriptives

Table 9 reports, for each country in the analysis sample, average government spending and average government revenue, both in percent of GDP, over 1996–2019, ranked and grouped by QoG rather than region, as in Table 8.

Table 9: Average government spending and revenue by country, 1996–2019

	Country	QoG	Spending (% GDP)	Revenue (% GDP)
	Paraguay	−0.83	15.58	17.98
	Ecuador	−0.74	28.97	28.63
	Guatemala	−0.72	11.96	12.50
	Dominican Republic	−0.52	14.42	13.89
	Argentina	−0.30	29.28	28.53
	Colombia	−0.29	23.62	27.26
	Peru	−0.27	18.29	19.60
Low QoG	Mexico	−0.20	20.42	20.66
	Brazil	−0.11	31.70	38.14
	Jamaica	−0.03	19.61	26.26
	Costa Rica	0.51	23.88	22.47
	Italy	0.55	43.64	45.95
	Uruguay	0.74	25.44	26.49
	Portugal	1.01	41.73	40.89
	Spain	1.14	39.19	38.32
	Japan	1.24	34.75	30.33
	Chile	1.27	22.18	23.26
	France	1.33	52.06	51.08
	Belgium	1.41	47.68	49.95
	United States	1.51	33.22	31.31
	Ireland	1.54	34.02	33.47
	Germany	1.68	43.81	44.90
High QoG	United Kingdom	1.69	39.65	38.58
	Austria	1.71	48.61	49.40
	Australia	1.76	34.17	34.46
	Canada	1.79	36.26	40.71
	Netherlands	1.90	43.63	44.85
	Sweden	1.91	49.96	52.12
	Denmark	2.01	51.41	55.11
	Finland	2.04	50.84	52.97

Source: Own elaboration based on data from Gethin (2023) and WGI.

Table 10 lists the definition and original data source of each of the variables used in the empirical analysis (Section 4.1).

Table 10: Variable definitions and sources

Variable	Definition	Source
Market Gini	Gini index of inequality in equivalized (square-root-scale) household market, pre-tax and pre-transfer, income	SWIID (Solt, 2020)
Disposable Gini	Gini index of inequality in equivalized household disposable, post-tax and post-transfer, income	SWIID (Solt, 2020)
Absolute redistribution	gini_mkt − gini_disp	SWIID (Solt, 2020)
Relative redistribution	(gini_mkt − gini_disp)/gini_mkt × 100	SWIID (Solt, 2020)
Total adjustment (FC)	Net size of the fiscal consolidation, % of GDP, zero outside consolidation episodes	Adler et al. (2024)
QoG	Simple average of the four WGI components below	WGI (World Bank, 2025)
Control of corruption	WGI control-of-corruption estimate	WGI (World Bank, 2025)
Government effectiveness	WGI government-effectiveness estimate	WGI (World Bank, 2025)
Rule of law	WGI rule-of-law estimate	WGI (World Bank, 2025)
Regulatory quality	WGI regulatory-quality estimate	WGI (World Bank, 2025)
Log real GDP per capita	Log of real GDP per capita, in millions of 2017 chained PPP USD (PWT rgdpo divided by raw, unscaled population)	Penn World Table (Feenstra et al., 2015)
Unemployment	Unemployment rate, % of labor force	World Development Indicators
Inflation	Inflation rate, % year-over-year	World Development Indicators
Trade openness	Exports plus imports, % of GDP	World Development Indicators
Government debt	General government gross debt, % of GDP	Fiscal-space database (Kose et al., 2022)
Urban population	Urban population, % of total population	World Development Indicators

Source: Own elaboration.